MILITARY SPECIFICATION

COMMUNICATIONS CENTRAL AN/TSC-99

This specification is approved for use by the Communications - Electronics Command, Department of the Army, and is available for use by all Departments and Angencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for Communications Central AN/TSC-99 which is a transportable radio communication station operating primarily in the HF radio frequency spectrum and which is designed to be used as a base station in a burst communication system.

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications and Standards. Unless otherwise specified, the following specifictions and standards of the issue listed in that issue of the Department of Defense Index of Specifictions and Standards (DODISS) specified in the solicitation, form a part of this specifications to the extent specified herein.

SPECIFICATION	
FEDERAL	
AA-C-00275	Chairs, Rotary and Straight Aluminum Office
MILITARY	
MIL-E-4158	Electronic Equipment, Ground, General Requirements for
MIL-W-6858	Welding, Resistance; Aluminum Magnesium, Non-hardening steels or alloys et al.
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-P-11268	Parts, Materials and Processes used in Electronic Communication Equipment

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Communications-Electronics Command and Fort Monmouth, ATTN: DRSEL-ED-TO, Fort Monmouth, New Jersey 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5820

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MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-F-15733	Filters, Radio Interference, General Specification for
MIL-I-23053/5	Insulation Sleeving, Electrical, Heat Shrinkable
MIL-W-45205	Welding, Gas Metal-Arc and Gas Tungsten Arc, Aluminum Alloys
MIL-E-45782	Electrical Wiring, Procedures for
MIL-S-55286	Shelter, Electrical Equipment S-280/G

STANDARDS

FED-STD-595	Colors
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attribute
MIL-STD-189	Racks, Electrical Equipment
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and other Devices
MIL-STD-285	Attenuation Measurements for Inclosures, Electromagnetic Shielding, for Electronic Test Purposes, Methods Of.
MIL-STD-454	Standard Requirements for Electronic Equipment
MIL-STD-461	Electronic Interference Characteristics, Requirement for Equipments
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-785	Requirements for Reliability Program
MIL-STD-810	Environmental Test Methods
MIL-STD-1332	Definition of Tactical, Prime, Precise and Utility Terminologies for Classification of the DOD Electric Power Engine Generator Set Family
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities

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MIL-STD-1474	Noise Limits for Army Material
MIL-STD-1857	Grounding, Bonding, and Shielding Design Practices

2.1.2 Other Government documents, drawings, and publications. The following government documents forma part of this specifications to the extent specified herein.

Specifications:

US Army CECOM Specification, CO-P-3200-AT Digital Message Device Group 8990()/P

US Air Force Specification, Bulletin 518a Cargo, Aircraft, Compartment Dimensional Data

Drawings:

SCC-16495	Binding Post U-106/U
DL-209947	Decontamination Port Drawing List
ESD-209948	Decontamination Port Assembly
SCD-539667	Rivet Nuts
SCD-539681	Cover Assembly
SCD-539711	Typical Entrance Box Installation
DL-621051	RFI Mod Kit MK-1079/G Drawing List
SCD-681010	Signal Entrance Box Assembly
SCD-681011	Entrance Box Welded Assembly
SCD-681075	Rivet Nuts
SCD-681076	Rivet Nuts
SCD-681080	Rivet Nuts

(Copies of these specifications and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Publications: The following documents are classified but are available for review at CECOM, Fort Monmouth, NJ upon request through the Contracting Officer for those having proper security clearances on file at CECOM Security Office:

MIL-HDBK-232 Military Standardization Handbook, RED/BLACK Engineering Installation Guidelines (title UNCLASSIFIED; document CONFIDENTIAL).

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NACSEM 5100, Compromising Emanations Laboratory Test Standard, Electromagnetics (Title UNCLASSIFIED, document CONFIDENTIAL).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein.

US Army CECOM Special Packaging Instructions SPI 1G00415
National Electrical Code NFPA 70-1978

American National Standards Institute ANSI Z35.1-1972

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 PRECEDENCE. This specification and the requirements contained herein take precedence over any referenced documents. In areas where the requirements do not cover a subject, and that subject is not specified elsewhere, the specifications covering that area shall be selected from the following precedence:

a. Military standards and specifications

b. Federal standards and specifications

c. Commercial standards and specifications (for example, ASA, ASNI, IEE, ASME, etc.)

d. In-house contractor specifications and standards

3. REQUIREMENTS

3.1 General. The Special Forces Burst Communication System provides voice and data communications between Army Special Forces base stations located in friendly territory and air dropped outstation detachments operating in hostile areas. The system uses single sideband emission in the 2-30 MHz HF frequency band as the primary mode of communication and satellite communication for special missions. The outstations will generally use the man-packed HF Radio Set AN/PRC-70 and may also be equipped with Radio Set AN/PSC-3 for satellite communication. The HF radio message traffic employs the burst mode at approximately 300 words per minute in the ASCII code as used in the Digital Message Device Group OA-8990 ( )/P. The base station, Communication Central AN/TSC-99, uses the central and local processing units to perform the functions of a digital message device group.

3.2 Description. Communication Central AN/TSC-99 is a transportable station operating in the HF radio frequency spectrum and having many operations controlled by a processor. It consists of the following major components:

Transmitter Group OT-118/TSC-99

Receiver Group OR-218/TSC-99

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Receiver Group OR-218/TSC-99

Antenna Systems

Engine generators (used with but not a part of the communication central)

Load Bank for engine generators (used with but not a part of engine generator)

Telephone Wire WD-1/TT (user supplied)

The transmitter and receiver groups are housed in separate shelters mounted on mobilizers and may be separated at distances up to five miles. The shelters are connected by field telephone wire for the message and control circuits.

3.2.1 Transmitter Group. The transmitter group contains four SSB, 1kw, HF transmitters and a local control processor which is subsidiary to the central processor in the receiver group. Two transmitters and associated antennas will be used for the normal flow of traffic while the third will be used for special missions and the fourth will be a spare but connected for service. All four may be operating at the same time. The transmitter functions are remotely controlled from the receiver group over voice grade wire lines to permit unattended operation. The local control processor operates the transmitters, monitors their performance and initiates switching functions upon command of the central processor. The processor also has facilities for sending burst transmissions from the transmitter shelter using a paper tape input punched in Baudot code in case of intershelter wire failure.

3.2.2 Receiver Group. The receiver group functions as a radio net control station, a message relay between radio and teletype wire circuits, and a remote control facility for the transmitter group. There are four HF receivers which may be connected as two space diversity terminals (primary use) or four independent receiving terminals (alternate use). Each receiver operates under the control of the central processor in either a frequency scanning mode for non-scheduled traffic or a time-frequency mode for scheduled traffic. The central processor controls the transmitters and automatically sends stored outgoing messages in the burst mode in accordance with instructions entered by keyboard, or by message tape header for time of transmission, transmitter frequency, transmitter power, and antenna. The processor performs the digital code and data rate conversion between the burst and teletype modes for both incoming and outgoing messages.

Facilities are provided for slow speed CW message traffic using magnetic tape recorders and off-line playback equipment for copying incoming messages and a teletype-to-morse converter for sending outgoing messages from punched tape. The off-line COMSEC equipment requires the use of punched paper tape in teletype code resulting in hand transfer of incoming and outgoing message tapes. An HF spectrum monitor is included to assist in the selection of outstation frequency assignments together with the long range prediction data stored in the central processor. A

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communications plan.

A 5th receiver, controlled thru the processor keyboard, normally monitors a single emergency frequency, and its output is patched to a magnetic tape recorder to receive nonscheduled burst, CW, and voice messages. A 6th receiver is a spare, stored in the receiver group, and is available to replace any defective on-line receiver.

3.2.3 Antennas. The base station will be used with the following antennas which will be procured separately.

2 ea Sloping, wire curtain, log periodic antenna, Antena Group OE-317/TSC-99. All will be assigned to the Transmitter Group and will have a 3KW balun.

2 ea Rosette array antenna system, Antenna Group OE-316/TSC-99 (Hermes Electronic Ltd. Model 3R/4E26C) installed for space diversity reception. All assigned to Receiver Group. Each OE-316/TSC-99 includes:

1 ea Antenna Array 3R/4E26C

3 ea Power Supply/Multicouplers

1 ea Multi-Multicoupler

1 ea Combiner

6 ea Antenna Cables, RG-213; each 500 ft long

NOTE: Additional cables, RG-58; are required but not supplied for multicoupler and combiner connections as given in paragraph on antenna multicoupler assembly.

3.2.4 RF Transmission line. Coaxial cable transmission lines shall be provided except as noted below, to connect the antennas to the shelters in lengths of 500 ft. each. The cables shall be supplied on reels.

(a) Receiving Antenna Cable. Six 500 ft length cables, consisting of RG-213/U, are supplied as part of each Antenna Group OE-316/TSC-99. These cables have the UHF Series Connector PL-259 at the end which connects to the receiver group shelter.

(b) Receiving Antenna Cable. One 500 ft length cable, consisting of RG-213/U with connector PL-259 at each end, shall be provided by the contractor together with a reel. This cable is intended for use with a dipole or other antenna.

(c) Transmitting Antenna Cable. Three 500 ft cables of RG-218/U shall be provided for connection between the shelter and Antenna Group OE-317/TSC-99. This length shall be provided in two 250 ft sections with each section on a reel. Each section

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Antenna Group OE-317/TSC-99. This length shall be provided in two 250 ft sections with each section on a reel. Each section shall have a male Type LC Series connector on each end. LC Series adapter shall be provided to connect two sections to form a 500 ft long cable.

(d) Adaptors. Two adaptors shall be provided to connect the Type LC Series male connector on the cable to a 7/8" EIA flange on the balun of Antenna Group OE-317/TSC-99.

(e) Satellite Antenna Cable. One 12 1/2 ft cable, consisting of RG-214/U with Type N connectors, is supplied with the antenna system of Radio Set AN/VSC-7.

3.2.4.1 Cable Grip. A cable grip shall be supplied for the shelter end of each transmission line with the exception of the satellite antenna cable for the purpose of preventing strain of the antenna entrance panel connector. The Cable Grip shall be the woven wire, side opening type, part number EQA37-H as made by Economy Cable Grip Co., South Norwalk, Conn., or similar. The transmission lines to be supplied with cable grips include those of the loop arrays, Antenna Group OE-316/TSC-99, in quantities of six for each antenna group.

3.2.5 Inter Shelter Wire: Field telephone Wire WD-1/TT will be used between the receiver and transmitter shelters for communication and data transmission. This wire consists of two insulated conductors each having 3 steel strands and 4 copper strands. One pair will be used for each circuit and the length may extend for 5 miles maximum. The wire will be supplied by the user of the communication central and is not a deliverable item on the contract.

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3.2.6 Engine Generators. It is planned to use the Electric Power Plant AN/MJQ-10 with each shelter of the Communication Central AN/TSC-99 but it will not be a part of the station. The AN/MJQ-10 consists of the following:

2 ea 30 kw Diesel Engine Generators mounted on individual trailers.

1 ea Switch Box for automatic paralleling of generators and for switching the load from one to the other generator.

2 ea Reels of Power Cable each 11 feet long for connection between generator and switch box.

3.3 Transmitter Group. The transmitter group shall be an operational assembly of the major items listed below plus ancillary equipment listed elsewhere in this document and forming a functional system in accordance with the block diagram, figure 1. The equipment shall be installed in Shelter, Electrical Equipment, S-280B/G, in accordance with the shelter requirements of this specification. The equipment will be a combination of standard military items and commercial items which have a history of good reliability. In general, the equipment will operate unattended but it shall be configured on the basis of human factors for ease of maintenance. The final configuration shall be a joint decision of the contractor and the Government based on MIL-STD-1472, Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.

3.3.1 Major Components.

4 ea - 1 kw Transmitter with power supply and exciter. (3 in use intermittently, 1 spare)

1 ea - Antenna Switch Matrix (remote controlled)

1 ea - Local control processor subsidiary to remote central processor

1 ea - Keyboard/printer (for processor control/readout)

1 ea - Modem (for outgoing radio messages)

2 ea - Processor modems (for communication with receiver group processor)

1 ea - Patch Panel for message circuits

6 ea - Patch Cords

1 ea - Message line matrix (remote controlled)

1 ea - HF receiver

1 ea - Punched paper tape reader

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2 ea - Power transformers (1 three phase, 1 single phase) for connection of Transmitter Group to any commercial power system)

2 ea - Transformers, 115/230 Volt, for air conditioners

2 ea - Whip Antenna (1 in use and 1 spare) (GFP)

4 ea - Audio line amplifiers (for message lines from receiver group)

1 ea - Intercom Equipment

1 ea - Dummy Load (for transmitter)

1 ea - Microphone

1 ea - Headset

1 ea - Manual telegraph key

1 ea - 1 kHz Tone oscillator for manual telegraph

1 ea - Test Equipment - Multimeter AN/USM-223 or equivalent (GFP)

1 ea - Telephone TA-312/PT (2 wire) (GFP)

1 ea - Storage cabinet for instruction books and small parts

1 ea - Table/bench

2 ea - Chairs

1 ea - Rechargeable battery power (local processor emergency power)

1 ea - Shelter Power Cable (100 ft.)

3.3.1.1 The following items shall be commercially available models:

Antenna Switch Matrix
Antenna Switch Matrix Control
Modems
Patch Panel & Cords
Audio Line Amplifiers
Mounting Shelf
Power Supply
Oscillator
Intercom Equipment
Processors
Dummy Load
Microphone
Headset
Radio Interface Assembly
Message Line Matrix

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Telegraph key
Uninterruptible Power Supply
Batteries for Uninterruptible Power Supply
Receiver
Paper Tape Reader
Intermediate Distribution Frame
Keyboard/Printer
Power Transformers (1 and 3 phase)
Clock
Gun Rack
Chairs
Peripheral Switch
Peripheral Switch Controller and Power Module
Mag Tape Unit and Mag Tapes
Power Supply for Mag Tape
Racks
Transformers (Air Conditioner)
Storage Cabinets
Transmitters
Exhaust Blower
Power Distribution
Storage Drawers
Mounting Shelves
Lights
Plug Mold

3.3.2 Required Transmitter (including exciter) Characteristics

RF Power output - High: 1 kw PEP, and 1 kw average

Low: 400 watts PEP/AVE ± 25%

Output Impedance - - - - 50 ohms unbalanced

Load VSWR: - - - - - - - Capable of operation up to 3:1 and protected from damage above 3:1 VSWR

Frequency Range - - - - 2.0000-29.9999 MHz in 100 Hz steps

Frequency Tuning - - - Front Panel manual control and remote control by a processor

Frequency Stability ----- 1 part in 10^{-8} per day

Required emission - - - USB and AME on USB

NOTE: Manual CW employed by this system is ON-OFF keying of 1 kHz tone which modulates transmitter in SSB mode. Burst keying employed by this system is audio frequency shift keying (FSK) which modulates transmitter in SSB mode.

Harmonic Suppression - - At least 55 dB below fundamental for second

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At least 55 dB below fundamental for third

At least 70 dB below fundamental for fourth and higher including all spurious emissions

Audio Input Impedance - 600 ohms balanced

Audio Input Level - - - 0 dBm for full rated output

Control - - - - - - - - - By a local processor in transmitter shelter plus manual control.

Temperature - - - - - - Operation = +32° to +120° F. Storage -50° to +155° F.

Humidity - - - - - - - - - 0 to 95%

3.3.3 Local Transmitter Processor Functions. The transmitter and antenna switching shall be controlled by a local transmitter processor with a duplicate stand by backup which is capable of performing the following functions on command from a central processor in the receiver group and on command from local keyboard:

Set transmitter frequency

Set transmitter output power (High or Low)

Select antennas or dummy load

Perform routine maintenance checks

Accomplish fault isolation and notify the central processor in the receiver group.

Readout routine maintenance information to either local printer or to receiver group printer upon command.

Readout traffic summary data to receiver group printer upon command.

Connect any message line to any transmitter upon command of the central processor. A switching system shall have the facility for connecting 4 separate message lines to separate transmitters for simultaneous transmission of traffic as well as connecting one message line to 1 or 2 or 3 transmitters for simultaneous transmission on different frequencies.

Provide message handling capability for transmitting burst messages from a punched tape reader located in the transmitter shelter. The transmitter commands for the messages shall be entered into the local processor from the keyboard or from the tape header. The processor shall perform the Baudot to ASCII code conversion for the messages, store messages, and send them in the burst mode. This facility is intended to be used when

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there is a failure of wire circuits from the receiver group. The local processor shall send messages in the burst mode at the scheduled time, as entered by keyboard. These functions shall be performed by a local processor located in the transmit shelter.

3.3.4 Local Processor Requirements. The detailed requirements for the processor and I/O devices located in transmit shelter, duplicate those for the central processor in the receiver group which are given further on in this document. However, the message handling requirement is limited to outgoing message traffic. Provision shall be made in the local processor for the future addition of two I/O devices to the following list. A duplicate standby processor is required.

3.3.4.1 Tape Reader. A punched paper tape reader shall be provided to read message tapes into the local processor for burst transmission from the transmitter group. This will be used when land lines from the receiver group become inoperative. Provisions shall be made to convert the message handling to minicartridge magnetic tape in place of punched paper tape at some future time.

3.3.4.2 Page Printer. A page printer shall be provided to print out processor instructions, traffic summaries, diagnostics, etc.

3.3.4.3 Keyboard. A keyboard shall be provided to control the local processor from the transmitter group for local message handling and maintenance routines.

3.3.4.4 Magnetic Tape Cassette Reader. Facilities shall be provided to use magnetic tape cassettes to read in programming information. Programming cassettes shall be provided in duplicate and have lock-out provision to avoid inadvertent erasures. Two blank cassettes shall be provided for use during diagnostic testing.

3.3.5 Clock. A mechanical clock (See paragraph 3.13) shall be mounted on the wall of the transmit shelter.

3.3.6 Manual Control. The control system in the transmitter group shall permit manual operation of the transmitter without using processor control as required in the case of maintenance or processor failure. It shall be possible for the transmitter operator to patch around the message line switch matrix and manually operate the antenna switch matrix without the services of the local processor.

3.3.7 Antenna Switch Matrix. An antenna switch matrix shall be provided which will connect any transmitter to any antenna upon command of the local processor in conjunction with commands from the central processor. The antenna switch matrix shall have sufficient switching capacity to connect the 4 transmitters to any of the 6 antenna terminals and to a dummy load and permit simultaneous operation of all transmitters after connections are established. The matrix shall have indicators to show how the inputs and outputs are connected.

(a). There shall be provision for manual operation of the matrix from a control panel in the event of processor failure.

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(b). Suitable interlocking shall be provided to prevent transmitter operation during switching or absence of antenna.

3.3.8 Transmitter Dummy Load. One fifty (50) ohm dummy load shall be installed which shall be capable of continuous duty operation with an RF power input of 1000 watts. VSWR over the frequency range of 2 to 30 MHx shall not exceed 1.1 to 1. The dummy load shall be connected to the antenna switching matrix.

3.3.9 Line Patch Panel. A patch panel with six patch cords shall be provided to permit manual connection of the telegraph tone key line and the message section from the local processor, to any of the four message line matrix inputs. Four incoming message lines shall be normalled through the patch panel to the message line matrix. The spare terminals in the signal entrance panel shall be connected to jacks on the patch panel and labeled spares.

3.3.10 Message Line Matrix. A switching matrix shall be provided which will permit connection of any of four incoming message lines to any of the four transmitters upon command of the local processor. In addition the switch facilities shall permit any one message line to be connected to any two transmitters for simultaneous transmission of traffic on two different radio frequencies. Indicators shall be provided to show which inputs and outputs are connected.

3.3.11 Line Amplifier. A line amplifier shall be provided for each of four message lines as shown in the block diagram in the appendix to compensate for wire line losses. Provision shall be made for monitoring and adjusting the levels of incoming voice frequency message signals intended for modulating the transmitter. Control of signal levels in the range of -36 dBm to +10 dBm is required. A meter on the transmitter which provides signal level information shall be used for adjusting the levels.

3.3.12 Telegraph Key, Microphone and Headset. The transmitter group shall be equipped with a telegraph key, Microphone and Headset. The telegraph key shall key a 1 KHz tone for modulating the transmitter. There shall be provisions for listening to the side tone. Connection to the audio input of the transmitter shall be made by patch cord at the message line patch panel. A suitable commercial type key, microphone and headset shall be provided. The microphone shall be a handheld type.

3.3.13 Processor Modems. Modems shall be provided for duplicate two-way communication between the central processor in the receiver group and the local processor in the transmitter group over wire lines for processor control service. These modems shall interface with the central processor control modems in the receiver group. The modems shall contain a line amplifier capable of compensating for line losses of up to 5 miles of field telephone wire. The amplifier shall have a level indicator to enable the level to be set properly.

3.3.14 Message Modem. A message modem, duplicating the type used in the receiver group, shall be provided for sending burst messages from the transmitter group.

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3.3.15 Emergency Power. A battery power system shall be provided in accordance with paragraph 3.15.11 which will keep the processor energized for 1 hour when primary power fails.

3.3.16 Telephone TA-312/PT. The field Telephone TA-312/PT (GFP) shall be provided for communication with other areas and stowed in the shelter. A two-wire line shall be provided from one pair of binding posts on the signal entrance panel to suitably located binding posts in the shelter.

3.3.17 Intercom Equipment. One terminal of a two terminal intercom system using loudspeakers shall be installed in the transmitter shelter for communication over a 2 wire line with the receiver shelter. A talk-listen switch may be used. The emergency battery power system shall be used to power the intercom to insure operation in case of loss of shelter primary power.

3.3.18 Monitor Receiver and Speaker. One radio receiver with loudspeaker shall be provided for monitoring the transmitters. The receiver shall be the same type used in the receiver group but operated manually. Provision shall be made for connection of the antenna terminals of the receiver to an outside whip antenna. Transmitter signal monitoring within the shelter shall be possible without the whip antenna being connected.

3.3.19 Whip Antenna. A whip antenna shall be provided for the monitor receiver in the transmitter shelter. The (GFP) antenna shall consist of Mast Sections MS-116, MS-117, and MS-118 and Mast Base AB-15/GR installed on a bracket on the door end of the shelter. Two sets of antenna sections (one as spare) shall be included and stored in the transmitter shelter.

3.3.20 Antenna Entrance Panel. An antenna entrance panel shall be provided which will afford exterior facilities for connecting the transmission lines from six antenna systems. The entrance panel shall allow sufficient space adjacent to each connector to permit easy attachment and safe removal of the R.F. coaxial connectors. On the interior side of the entrance panel, RF cables shall connect with the antenna switch matrix.

3.3.21 Signal Entrance Panel. A signal entrance panel shall be provided for the connection of the following external signal and control wire lines in accordance with 3.6.3, Entrance Panels and Cover Assemblies.

1 pair - Telephone Line for TA-312/PT

1 pair - Intercom line to receiver group

4 pair - Control lines to receiver group for duplex transmissions between the processors

4 pair - Message lines to receiver group

3 pair - Spare

Two (2) binding posts shall be provided for each line. RFI line filters in

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accordance with MIL-F-15733 shall be provided in each line. Electronic surge protectors shall be provided in each line between the filters and binding posts of the signal entrance panel.

3.4 Receiver Group. The receiver group shall be an operational assembly of major items listed in the following paragraph plus ancillary equipment listed in this document and forming a functional system in accordance with block diagram, figure 2. The equipment shall be installed in a lengthened shelter, Type S-280C/G in accordance with shelter requirements of this specification. The equipment shall be a combination of standard military items and commercial items which have a history of good reliability. The final configuration shall be a joint decision of the contractor and the government based on MIL-STD-1472, Human Engineering Design Criteria for Military Systems, Equipment, and Facilities. An objective is to operate with 4 operators having duties as follows:

(a) Receiver Operator.

1. Operates receiver when in manual mode.

2. Operates magnetic tape recorders when receiving CW or voice traffic.

3. Operates receivers and transmitters, using keyboard control, when receiving and sending voice and manual CW.

4. Maintains incoming radio message log.

5. Transcribes CW messages from the magnetic tape recorders into the processor from the keyboard. Edits messages, adds header information, and has the processor punch a tape of the message.

6. Monitors the frequency spectrum monitor.

7. Inputs long-range frequency prediction data and selects outstation frequencies.

8. Prepares frequency assignment messages to the outstations which are processed by the COMSEC operator into punched tape and sent in the same manner as other outgoing radio messages.

(b) System Controller.

1. Inputs incoming and outgoing message traffic schedules by keyboard command as required.

2. Monitors and maintains system status and configuration tables.

3. Coordinates system equipment usage activities.

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4. Recalls traffic data from CPU for the daily report to headquarters.

5. Initiates diagnostic routines upon indications of trouble in the transmitter group.

(c) COMSEC Operators (Normally two).

1. Processes all messages (on punched tape) from both radio and Tactical Operations Center (T.O.C.)

2. Prepares dummy messages (on punched tape).

3. Monitor tape punch when operating under CPU control. Mark message tapes with header identification.

4. Input outgoing message traffic (from punched tape) into the central processor together with transmitter operating instructions and message sending time. Inputs data rate instructions for 300 or 1200 bits per second when outgoing message traffic is routed over the satellite communication circuit.

3.4.1 Major Components.

1 ea Central Processor including duplicate backup with the following:

2 ea Punched Paper Tape Reader - (For message traffic BAUDOT code)

3 ea Paper Tape Punches - (1 for message traffic BAUDOT code)
(1 for daily traffic summaries)
(1 for spare but connected for Service)

2 ea Magnetic tape cassettes reader/recorder - (For CPU loading and archive functions)

2 ea Keyboard & Display Units (For CPU controller & receiver operation)

2 ea Page Printer (For activity log & processor information)

5 ea Message Modems (Includes demodulator)(4 in use, 1 spare)

2 ea Processor Modems - (for communication with transmitter group processor).

6 ea HF Radio Receiver (5 in use, 1 spare)

2 ea Satellite Radio Set AN/VSC-7 (A future addition) (GFP)
(1 in use, 1 spare)

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2 ea Antenna Multicoupler Assembly (for rosette antennas) (GFP)

1 ea Antenna Multicoupler (for other antenna)

4 ea Magnetic Tape Cassette Recorder/Reproducer, Audio
(2 used for recording, 1 for playback, and 1 spare)

1 ea Shelter Power Cable (100 ft.) on a reel.

2 ea - Power Transformers (1 three phase, 1 single phase) (For connection of Receiver Group to any commercial power system)

2 ea - Transformers, 115/230 volt, for air conditioners

10 ea Magnetic Tape Cassettes (for CW and voice message traffic)

2 ea Teletype to Morse Converter (1 in use, 1 spare)

1 ea Patch Panel

1 ea HF Spectrum Monitor (R-2093/TRQ-35(V)) (User installed)

1 ea Secure Teletype Facility (incl 2 ea KW-7/TSEC &
1 ea KWX-11/TSEC w/o Power Supply which are user
installed. Power supply provided by and
installed by contractor)

1 ea Telegraph Terminal TH-22/TG (GFP)

2 ea COMSEC Terminal AN/FYQ-90(V)1 with Reader/Punch RD-486/U

1 ea Telephone TA-938/G (GFP)

1 ea Telephone TA-312/PT (GFP)

1 ea Intercom

3 ea Microphone (2 in use, 1 spare)

3 ea Monaural headset, 600 Ohms (2 in use, 1 spare)

2 ea Manual Telegraph Keys

1 ea Emergency battery power system

1 ea Multimeter AN/USM-223 or equivalent (GFP)

1 ea Table/bench

1 ea Security container (two drawer with combination lock)

4 ea Chairs

Military Versions
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3.4.1.1 Commercial Items.

The following shall be commercially available models:

MODEMS
PATCH PANEL & CORDS
INTERCOM EQUIPMENT
MICROPHONE
HEADSET
TELEGRAPH KEY
UNINTERRUPTIBLE POWER SUPPLY
BATTERIES FOR UPS
RECEIVERS

PAPER TAPE READER
PAPER TAPE PUNCH
INTERMEDIATE DISTRIBUTION FRAME

MASS MEMORY AND CONTROLLER
AUDIO RECORDERS AND CONTROL
AUDIO CASSETTES
GUN RACK
BULLETIN BOARD
CHAIRS
FIRST AID KITS
VENT FANS
SHOCK ISOLATORS
STORAGE CABINETS
RADIO INTERFACE ASSEMBLY

TTY-MORSE CONVERTERS
READER/PUNCH (TEMPEST CERT)
AUDIO PANEL
PROCESSORS
I/O CHASSIS
MAG TAPE UNIT
MAG TAPES
POWER SUPPLY FOR MAG TAPE
PAGE PRINTER WITH TAKE-UP DEVICE
KEYBOARD/DISPLAY
RACKS
TRANSFORMERS, POWER (1 and 3 phase)
POWER DISTRIBUTION PANEL
ANTENNA MULTICOUPLER
STORAGE DRAWERS
MOUNTING SHELVES
CABLE RACEWAYS
LIGHTS
PLUG MOLD
CLOCK
COMSEC FACILITIES
TRANSFORMERS, air conditioners
POWER SUPPLY FOR KWX-11

3.4.2 RECEIVER. The receivers shall be commercially available models meeting the essential requirements listed below. The receivers may have additional features and characteristics which are generally found in commercial models but must meet the essential requirements for system operation of this communication central. Five receivers shall be connected for service and one available as a replacement spare.

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Frequency Range: 2.0000-29.9999 MHz in 100 Hz increments

Frequency Stability: 1 part in 10^{-8} per day.

Frequency tuning: (a) Manual control at front panel.
(b) Remote control by a processor for operation in (1) time-frequency mode and (2) frequency scanning mode.

Receive Modes: CW, AM, & USB

Sensitivity (SSB): 0.35 uv for 10 dB (s+n)/n

Bandwidth SSB: 2.7 kHz nominal

Audio bandwidth: 300 - 3500 Hz nominal

Spurious Response: All receiver spurious response outside of ± 2.5% of the tuned frequency shall be down at least 60 dB.

Intermodulation: In band, -40 dB from either of two input tones, at 0.1 volt per tone. Out of band, 2nd and 3rd order, -80 dB from either of two input signals, 10 mV per tone, 50 kHz or more off center frequency.

Image Rejection: At least 100 dB

Antenna Input: 50 ohms impedance nominal. BNC type connector.

RF Overload Protection: Up to 100 volts RF input, power on or off, automatic reset.

Audio output: Line - - 0 dBm nominal, 600 ohms, balanced adjustable -20 to +10 dBm

Headset - 600 ohms nominal, level adjustable.

Speaker - 2 watts nominal to remote speaker, level adjustable.

Audio Hum and Noise: -40 dB on line audio output.

Remote Control: Facility for remote control of frequency by a processor. Facilities shall be included which will permit the processor to operate the receiver in a scanning mode covering up to 7 discrete frequencies and locking to a frequency when message traffic is present.

Primary Power: 115 VAC ± 10%, single phase, 50/60 Hz.

Temperature: Operation = +32° to +120° F. Storage -50° to +155° F.

Humidity: 0 to 95%.

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3.4.3 Message Modem. Five two-tone, FSK modems shall be provided to interface between the receivers and processor and to interface between the processor and transmitters for message service. Each modem shall contain a modulator, demodulator, space diversity combiner and decision threshold combiner. Four modems are required for system operation with a spare fifth unit on hand for manual replacement.

3.4.3.1 Burst Message Code. The message modems and the message section of the processor must be compatible with the burst message code used by the communication system. The bit structure of the characters and the data rate in the burst message is that used by the Digital Message Device Group OA-8990( )/P which is described in the specification listed in paragraph 2.2.4. The characteristics are summarized below:

A. Bit Rate 266.6, 300, & 1200 BPS

B. Character Length 6 Bits

C. Preamble 64 Bits

Includes recognition code which allows selection of up to 64 different addresses for message routing.

D. Postamble 64 Bits

E. Error Detection/Correction Scheme

Uses a 78/127 Cyclic Block Code with interleaving. Each 127 bit block consists of 78 information bits (13 characters) and 49 parity (check) bits.

The OA 8990 ( )/P HF output bit rate is 266.6 bps with tones spaced at 1303 Hz and 2103 Hz to achieve orthogonality. Error detection and correction with interleaving are employed by the OA-8890 ( )/P for both HF radio and satellite transmission in the burst communications systems.

3.4.3.2 Modulator. The modulator section of the modem shall be connected to the message output section of the central processor and shall produce two-tone phase continuous FSK in accordance with the burst transmission characteristic and data rate used in Digital Message Device Group OA-8990( )/P. The keyed tones will be sent over voice grade wire line to the transmitter shelter and used to modulate the transmitter.

3.4.3.3 Demodulator. The demodulator section of the modem shall be connected to the receiver output and convert the signals to a form suitable to input to the central processor. Separate matched filters centered around each tone and optimized for the data rate shall be used to detect the respective mark and space frequencies with each tone channel carrying all information on an on-off keying basis. Each demodulator shall contain provisions for combining inputs from another modem on a post detection integration basis to permit space diversity operation of the

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receiving system. Facilities shall be provided at a patch panel to connect the modems to operate as a space diversity pair or as non-diversity independent channels. Decision threshold logic shall be applied after the diversity combiner to combat multipath and approach another order of inband frequency diversity performance such that when a fade is detected in one channel, the other independent channel is used to make the decision.

3.4.4 Antenna Multicouplers.

3.4.4.1 Antenna Multicoupler Assembly. Installation facilities shall be provided for the antenna multicoupler equipment for two rosette arrays, (OE-316/TSC-99) (See Paragraph 3.2.3) and cables furnished for connections for omnidirectional operation in accordance with the block diagram, figure 3. The manufacturer's instructions shall be followed when making connections to the equipment in order to maintain the required phase relationship. The length of each cable within each of the groups listed below must be the same within ± 5 mm (0.2 inches):

(a) The group of 6 cables connecting the antenna with the antenna entrance panel (P/O OE-316/TSC-99).

(b) The group of 6 cables connecting the antenna entrance panel with the 3 power supply/multicouplers and 3 multicouplers (Contractor furnished).

(c) The group of 6 cables connecting 3 power supply/multicouplers and 3 multicouplers with the combiner (P/O OE-316/TSC-99).

Note: The cable length (P/O OE-316/TSC-99) between the combiner and final multicoupler is not critical. The length of cable (contractor furnished) to each receiver is not critical.

3.4.4.1.1 It is anticipated the the final receiver-multicoupler connections will be made to meet the directional requirements of the particular communication circuits when the base station is installed on site and will not be subject to change unless the location of the distant stations change substantially.

3.4.4.2 Antenna Multicoupler for Other Antennas. The contractor shall provide a multicoupler having a minimum of 4 receiver outputs. The receivers or spectrum monitor may be connected to this multicoupler at the time of installation as alternate to the array antenna. Cable connection shall be provided between the multicoupler and a spare terminal in the antenna entrance panel.

3.4.5 Satellite Communication Equipment. Two Satellite Base Stations AN/VSC-7 will be used in the receiver shelter. The one in use shall be connected with its "DATA" receptacle connected to the CPU message section (both incoming and outgoing). The send/receive switching function is operated automatically by the outgoing message preamble. The spare shall be connected to power but data and antenna connections omitted. The AN/VSC-7 handset shall also be connected to the "VOICE/DATA" or "RXMT" receptacles to permit push to talk voice operation when the AN/VSC-7 is

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manually switched to the patch panel to permit recording of incoming voice.

3.4.5.1 Satellite Base Station AN/VSC-7. Space shall be provided in the receive shelter to install 2 AN/VSC-7 base stations. A minimum vertical height or 8.75 inches and a depth of 20 inches is required for each base station in a standard 19 inch rack. Each AN/VSC-7 comes equipped with a 19 inch rack mount shelf. A standard 3 prong, 115 volt, single phase power outlet shall be provided in the immediate vicinity of each mounting location to provide primary power when the units are installed. The satellite signal cable for the unit in use shall be terminated with a connector compatible with the An/VSC-7 signal port.

3.4.5.2 Antenna. The antenna for the AN/VSC-7 is a crossed log periodic mounted on the end of a metal tube. Mounting hardware shall be provided on the shelter such that the antenna is removable during transit of the shelter. The mounting shall be located in the upper right of the door end of the receiver shelter. Necessary antenna mounting dimensions will be provided by the Government in a separate action.

3.4.5.2.1 Antenna Cable. An interior antenna cable shall be provided for the satellite base station in use. The cable shall be connected to the antenna entrance panel in the receive shelter and shall be of sufficient length to reach the space provided for mounting the unit in use plus a service loop. Coaxial cable RG-214 shall be used with Type N connectors on each end.

3.4.5.3 Satellite Preamble. When a transmission instruction routes an outgoing burst message via the satellite, the processor shall make the preamble for the AN/VSC-7 2.1 seconds long prior to transmission of message at 300 bits/second data rate and 500 milliseconds long prior to transmission of message at 1200 bits/second data rate. The preamble is referenced under Central Processor, paragraph 3.4.8.5.

3.4.6 HF Spectrum Monitor. Provision shall be made for the installation of an HF spectrum monitor, Barry Research Communications Co. Model RSS-4 (nomenclatured as radio Receiver R-2093/TRQ-35(V), in the receiver shelter to collect and display HF propagation information to the system controller. The RF input cable shall have sufficient length to enable it to be connected to any one of several antenna multicouplers. The cable connecting a multicoupler with the spectrum monitor shall have a Type N connector at the monitor end.

3.4.7 Patch Panel Facilities. Where patch panel facilities are provided for switching message channels (i.e., connecting modems in diversity or non-diversity), the patch panels shall be located so that the jack identification is readable to the seated receiver operator and the patch cords in use do not obscure indicators or interfere with the operation of controls of other equipment.

3.4.7.1 Patch Panel. A patch panel shall be provided to permit connection of magnetic tape recorders to the receiver outputs including the satellite radios for recording messages sent by manual CW and voice. The patching facilities shall provide for the connection of each channel of a two-channel recorder to separate receivers for space diversity

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reception as shown in the block diagram in the appendix. When patch cords are not installed, the outputs of four receivers and satellite radio shall be connected through to their demodulators. The fifth receiver shall also be connected to the patch panel so that it can be patched to any magnetic tape recorder.

3.4.7.2 Patch Cords. Seven patch cords (5 in use and 2 spares) shall be provided for the patch panel.

3.4.8 Central Processor. A central processor with a duplicate standby backup including memory shall be provided which will perform many of the base station's control and message handling functions on a program basis as outlined in the following paragraphs and as shown in the block diagram in the appendix. The computer equipment shall be selected in accordance with the government's recommended list and contract note. Any substitute must be fully justified based on technical, logistical, maintainability, and cost factors. Supportability over its life cycle is of prime importance for this equipment. Sample data formats are given in Appendix A.

3.4.8.1 Receiving Control Functions

(1) Set each of four HF receivers to a specific frequency at a specific time.

(2) Operate each of four HF receivers in a scanning mode covering up to 7 specific frequencies and holding to a frequency when messag traffic is detected.

(3) Operate the four receivers in (space diversity) pairs and with each pair performing under an independent program (either the scanning mode or time-frequency mode)

(4) The fifth receiver will be CPU controlled but will be used to monitor a single frequency. The frequencies may change for day and night.

3.4.8.2 Sending Control Functions

(1) Dispatch stored messages on a time schedule.

(2) Route messages over specific wire lines to transmitter shelter and to one or more transmitters. It shall be possible to send the same message traffic simultaneously over two different transmitter operating on different frequencies.

(3) Set transmitter frequency, power level and select antenna in conjunction with the local processor at the transmitting shelter.

(4) Receive and display transmitter status including standby/operate, power level, and faults. A visual status indicator is desired at the receiver operator's position and at the controller's position

(5) Transfer outgoing burst message traffic to another idle transmitter within 60 seconds after receipt of a fault indication

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for the transmitter in use. This action requires the processor to provide antenna, frequency, and power commands to the idle transmitter and retransmit the message.

3.4.8.3 Miscellaneous Station Control Functions

(1) Initiate diagnostic procedure through the transmitter group processor and print out results by keyboard command.

(2) Provide operational reports from the transmitter processor including time of operations and power level by keyboard command.

(3) Maintain station message logs and produce daily incoming and outgoing traffic summaries (see Appendix A) on punched tape for transmission by teletype.

(4) Stop automatic operation and change over to manual operation which consists of using keyboard to:

(a) Put a specific transmitter and antenna "on line".

(b) Set transmitter frequency, power, and status (standby or operate).

(c) Send burst message from the proper memory file by keyboard command.

(d) Set receivers to a desired frequency by keyboard command. In addition, it shall be possible to set receiver frequency by using receiver front panel controls.

(5) Send by keyboard command, a continuous test signal (such as a series of bits) for use in adjusting the line amplifiers.

(6) Punch a paper tape of any systems report or status display on keyboard command.

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3.4.8.4 Incoming Burst Message Handling

(1) Distinguish between message signals and noise pulses by utilizing the preamble transmitted by the Digital Message Device Group OA-8990( )/P.

(2) Store HF messages at incoming speed and forward to punch at paper tape punch speed. Storage capacity shall accommodate a maximum of four 200-word group messages per channel before erasure.

(3) Convert messages from the ASCII code to Baudot code before forwarding.

(4) Four independent incoming message channels to the processor shall be provided as shown in the block diagram in Figure 2. Each channel shall have the capability to sense and adjust to incoming message data rates of 300 or 1200 bits per second (used by satellite system) or the data rate established by the Digital Message Device Group OA-8990/P for HF radio communication.

(5) Display message from any channel on display unit by keyboard command.

(6) Maintain message log file from keyboard or paper tape entries which must be read out on punched tape daily. See example in Appendix A.

(7) Route all incoming message traffic to the single primary message punch and provide sufficient buffering of multiple incoming message channels. Provide automatic changeover from the primary to the second or third backup punches in the event of punch failures.

3.4.8.5 Outgoing Burst Message Handling

(1) Message entered into processor from a punched paper tape in Baudot code. Processor converts message to ASCII code as used in DMDG.

(2) Processor automatically adds a preamble, error detection and correction coding, and a postamble. Details are given in Digital Message Device Group specification CO-P-3200AT.

(3) Administrative data for each message is entered into the processor by keyboard or punched paper tape and printed out on the station activity log (see Appendix B) simultaneously as follows:

(a) Message identification (i.e. date time group)

(b) Destination (outstation) code (i.e. outstation number)

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(c) Routing directions through base station:

I. Landline number

II. Transmitter number

III. Frequency

IV. RF power level

V. Antenna number

(d) Release time (i.e. date time group)

(4) Processor stores all messages until transmission time.

(5) Processor sends message one to nine times at release time in accordance with keyboard entry or paper tape send instructions, at the burst rate, with a fixed delay time between bursts, and records the transmission events on the station activity log. The fixed delay shall be the worst case processing time between received messages, required by the out-station digital burst device which is 50 seconds. When burst message is routed via satellite communication system, the data rate shall be either 300 or 1200 bits per second in accordance with the instructions entered by keyboard or tape. The data rate for the HF radio system will be that established by the Digital Message Device Group OA-8990/P for HF radio communication which is 266.6 bits per second.

(6) It shall be possible for the keyboard to call up and display the message while it is in the file.

(7) The maximum quantity of outgoing messages for which storage capacity is required in receiver group is 32 two hundred word groups (5 characters per group) in a 24 hour period. The transmitting group shall store up to 10 outgoing messages. All messages will have a preamble/postamble format compatible with DMDG OA-8890 ( )/P.

(8) Maintain an outgoing message log file which will be read out daily on punched tape. See example in Appendix B.

3.4.8.6 HF Frequency Data Bank

(1) Store MUF-FOT-LUF propagation prediction data for up to 36 communication paths between base station and outstations. Information shall be entered by tape reader from Baudot punched tape.

(2) Store a quantity of up to 100 frequencies which are prohibited because of assignment to higher priority users. The information

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will be entered by keyboard.

(3) Processor facilities shall permit the operator to:

(a) Display stored propagation data by path (or outstation identification) and time by keyboard command.

(b) Flag any frequency chosen by the system controller as the result of viewing the spectrum monitor and the MUF-LUF data but which is listed in the prohibited or previously used file.

(4) Maintain a 60-day file on past frequency assignment to all outstations up to 36 in number and based on 2 assignments per day per outstation. The file shall include information on frequency, path, date and time used.

NOTE: Outstation frequencies are assigned by the base station and are used only once in a period of several weeks.

3.4.8.7 Standby Processor. A redundant processor shall be provided for the local transmitter processor and central processor to shift control between the failed and standby system. The shift to the standby system shall be automatic or manual at the option of the operator and shall be accomplished in such a manner as to minimize the degradation of outgoing or loss of incoming messages.

3.4.9 Receiver Group Central Processor I/O Devices. Input/output devices shall be provided as specified in the following paragraphs. All I/O devices shall use standard off-the-shelf interfaces developed by the computer manufacturer. Provision shall be made in the central processor for the future addition of up to four additional I/O devices.

3.4.9.1 Paper Tape Reader. Two paper, punched tape, readers shall be provided in the receiver group and shall be adaptable to either 5 level Baudot or 8 level ASCII code. The primary function of the first reader is to read outgoing messages (in Baudot code) into the message section of the central processor. Facilities shall be provided to connect the auxiliary reader, when required, to read in propagation data to the processor file or to send outgoing messages through the teletype-morse converter for the slow speed CW mode of transmission.

3.4.9.1.1 Provision shall be made in the input circuitry of the processor to convert the message handling to mini cartridge magnetic tape recording in place of punched paper tape at some future time.

3.4.9.2 Magnetic Tape Cassette Read-Write. Facilities shall be provided in the central processor to use magnetic tape cassettes to input the

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computer program. It will also be used to read out stored data in the processor files such as frequency propagation in order to preclude loss when it is necessary to run off-line processor maintenance routines.

3.4.9.2.1 Magnetic Tape Cassettes. Cassettes shall be provided as follows:

(a) Cassettes containing programming instructions shall be provided in duplicate. These cassettes shall have a lock-out provision to avoid inadvertent erasures.

(b) Cassettes in duplicate containing off-line maintenance programs will be included but will be furnished by the Government.

(c) Cassettes for temporary storage of processor file data shall be provided in quantity of 4.

3.4.9.3 Display/Keyboard Unit. Supervisory on-line control of the base station shall be provided by two display/keyboard units. One unit shall be located at the controller position and the other at the receiver operator position. Both keyboards shall permit control of all the message handling functions and system control functions in the central processor as outlined in the Central Processor, paragraph 3.4.8. The display shall include a symbolic/alphanumeric presentation for data entry. All data entered shall be error checked prior to storage. Errors shall be flagged. The display shall provide entry verification on a character by character basis. The display unit shall be military qualified such as Interstate Electronics Model PD-3000.

3.4.9.4 Paper Tape Punch. Three punches shall be provided to read out incoming messages and shall be adaptable to either 5 level Baudot or 8 level ASCII code. The punches shall be capable of accepting either paper or mylar backed tape. The minimum speed shall be 10 characters/sec. Appropriate tape guides shall be provided and tape residue shall be caught in an enclosed container. The punches shall meet the ambient noise requirements of para. 3.6.13.5. One punch will be used to output all of the 4 incoming message channels of the processor. A second punch will be used to read out daily traffic summaries. The third punch will be a back-up spare connected for service.

3.4.9.4.1 Provisions shall be made in the output circuitry of the processor to convert the message handling to mini cartridge magnetic tape recording in place of punched paper tape at some future time.

3.4.9.5 Page Printer. Two page printers are required to readout information from the processor. One unit shall be used exclusively for an activity log to record outgoing message traffic administrative data in accordance with paragraph 3.4.8.5 and shall automatically print out the required information therein at the time when the event occurs. The second unit shall be used for readout of miscellaneous processor information including programming data in accordance with keyboard command. The printer shall be a military qualified, line printer such as Miltope Corp Model HSP-3609-212A with a roller type, paper take-up device

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on top of the printer. An appropriate I/O card shall be provided in the processor to interface with the printer.

3.4.9.5.1 Paper for Page Printer. A quantity of 2 boxes fanfold paper (6000 sheets) shall be supplied with each communication central for initial operation. The paper shall be the pressure sensitive type required by the printer in the previous paragraph 3.4.9.5.

3.4.9.6 Processor Modems. Modems shall be provided in duplicate for two-way communication between the central processor in the receiver group and the local processor in the transmitter group over voice grade, wire lines for processor control service. These modems shall interface with the local processor control modems in the transmitter group. The modems shall contain a line amplifier capable of compensating for line losses of up to 5 miles of field telephone wire. The amplifier shall have a level indicator to enable the level to be set properly.

3.4.10 Digital Clock. A digital clock having a 24-hour indication shall be provided within view of the system controller's and receiver operator's position for use in message handling and shall operate in conjunction with the processor. This requirement may be met by displaying clock time on the display unit at both the controller and the receiver operator positions providing that the time is displayed continuously without lengthly interruption from other information displays.

3.4.11 Central and Local Processor Common Control Subsystems. Common control shall carry out the centralized and time shared function of operation and monitoring. A stored program concept of control organization shall be used. This shall not preclude the use of read only memories for storing those parts of the program subject only to infrequent and minor changes. The control sub-system shall be designed and organized to accommodate specialized or new features solely by means of stored program changes with modular expansion of the memory. Information once stored in the common control sub-system shall not be lost due to power failure.

3.4.12 Central and Local Processor Memories. Sufficient memory shall be provided to accomplish all requirements of the burst communication system. The memory shall be protected against loss in the event of system/component failure other than memory failure. Maintaining maximum growth capability to accommodate possible future system changes shall be a design goal.

3.4.13 Magnetic Tape Recording and Playback.

3.4.13.1 Magnetic Tape Recorder/Reproducer. Two 2-channel magnetic tape audio recorders shall be provided for recording manual CW messages being sent at 5 to 30 wpm and voice messages. Tape cassettes shall be used to permit playback on other recorder/reproducers. The front panel shall have controls for start, stop, and level adjust. The recorder will normally be used for recording the same message from separate receivers operated in a space diversity mode. One recorder/reproducer shall be provided for off-line manually transcribing messages recorded on equipment in the preceding paragraph. There shall be a front panel control to adjust the

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speed of playback. There shall be a start, stop, and reverse controls to enable replay of a section of tape. Each channel shall have a level control. Monaural headsets will be used with this equipment.

The intent of this requirement is to record voice and manual CW on an as needed basis rather than on a continuous basis.

3.4.13.2 Magnetic Tape Cassette. Ten magnetic tapes cassettes shall be supplied for use with the tape equipment in the preceding paragraphs.

3.4.14 Message Line Switching. A switching function shall be provided to route messages from the central processor over any one or more of the message wire lines connecting the receiver group with the transmitter group. The function may be incorporated into the central processor unit.

3.4.15 Emergency Power. Batteries with a charging system shall be provided in the receive shelter in accordance with paragraph 3.15.11 to immediately provide energy for the time intervals indicated below in the event of engine generator or commercial power failures.

a. Processor functions for one hour.
b. Wire line modems for one hour.
c. Processor message sending facilities for one hour.
d. Emergency lighting for one hour.

3.4.16 Off-Line Message Processing Facility. Two off-line message processing facilities shall be installed. This equipment consists of paper tape reader/punch RD-486/U and keyboard display set AN/FYQ-90(V)1. The applicable encryption printed circuit cards will be provided by and installed by the user. The Baudot teletypewriter code is used for both input and output message tapes.

3.4.17 Secure Teletype Communications Facility. A paper tape reader/punch, which will operate with one Signal Converter KWX-11/TSEC and two Coder KW-7/TSEC in full duplex, shall be provided and installed as a secure communications facility for communication with the Tactical Operations Center over landline. The equipment must meet TEMPEST requirements. The installation facilities shall accommodate the two KW-7/TSEC and one KWX-11/TSEC with power supply. The KW-7/TSEC are Government loaned items (when required) but a power supply for the KWX-11/TSEC shall be provided by the contractor.

3.4.18 Teletype Terminal Equipment. One GFP Telegraph Terminal TH-22/TG shall be installed to interface the DC teletype loop with the telephone lines and provide voice frequency telegraph signals which are required for transmission over voice grade wire lines. This item shall be connected to a 4-wire line for full duplex operation.

3.4.19 Telephone. One push button Telephone TA-938/G (GFP) shall be installed and connected to the signal entrance panel for 2-wire service connection to a commercial telephone system. In addition, a pair of wires shall be run from the signal entrance panel to a location in the vicinity of the above telephone for use at some future time for Telephone

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TA-312/PT. The terminals in the signal entrance panel shall be labeled "TEL 1 and TEL 2" to identify the pairs. The Telephone TA-312/PT will not be installed by the contractor but shall be stored in the shelter.

3.4.20 Intercom Equipment. One terminal of a two terminal intercom system using loudspeaker shall be installed in the receiver shelter for communication over 2-wire line to the transmitter shelter. The emergency battery power system be used to power the intercom to insure operation in case of loss of shelter primary power.

3.4.21 Microphone. Two microphone connections shall be provided, one at the receiver operator position and one at the controller's position, for SSB voice radio communication. The microphone circuit shall be connected to a patch panel which will provide connection to route the voice circuit over a wire line to a transmitter. A commercial type microphone shall be provided by the contractor. An amplifier shall be provided if necessary to raise level to that required for wire line input. The microphone shall be a hand held type.

3.4.22 Headset. A headset connection with a receiver selector switch shall be provided at the receiver operator position and at the system controller position for listening to any of the five receivers or for listening to the sidetone when sending CW. Monaural commercial headsets shall be provided by the contractor. The headsets will be used, also with he recorder/reproducer for transcribing messages.

3.4.23 Telegraph Key. Two commercial telegraph keys together with a 1 kHz tone oscillator shall be provided for transmitting tone CW on SSB, one located at receiver operator's position and one at controller's position. The output of the tone oscillator shall be connected through a patch panel for routing to a transmitter.

3.4.24 Teletype to Morse Converter. Two Teletype-to-Morse Converters shall be provided to enable messages to be sent from punched tape in Morse code at a rate adjustable between 5 and 30 words per minute. Both converters shall be connected for service. Connection between the outgoing message tape readers and the Teletype-to-Morse Converters shall be made at the patch panel for both the data and control circuits.

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3.4.25 Antenna Entrance Panel. An antenna entrance panel shall be provided which will afford exterior shelter facilities for connecting the transmission lines from the antennas described earlier in this specification. Fourteen (14) receptacles (includes one spare) shall be provided and spaced to provide easy attachment and safe removal of the RF coaxial connectors. Unless otherwise specified in the paragraph on antennas or in the bid request, the receptacles shall mate with coaxial cable connector PL-259/U except one for the satellite antenna which shall be Type N.

3.4.26 Signal Entrance Panel. A signal entrance panel shall be provided for connection of the following external signal and control lines in accordance with 3.6.3.

1 pair - Intercom with transmitter shelter

4 pair - Message lines to transmitter shelter

4 pair - Processor control lines to transmitter shelter

1 pair - 2 wire Telephone line for primary telephone

1 pair - 2 wire telephone line for spare telephone

2 pair - 4 wire Teletype line

3 pair - spare

Two binding posts shall be provided for each line. Line filters in each line shall be provided to meet requirements of MIL-F-15733 (RFI Filters). Electronic surge protectors shall be provided in each line between the filters and binding posts of the signal entrance panel.

3.4.27 Safe. A two-drawer security filing cabinet with combination lock shall be securely installed in the receiver shelter. The safe shall bear General Services Administration approved security container label affixed to the outside of the drawer.

3.4.28 Monitor Speaker. If the receivers, both HF and SATCOM, do not contain speakers, one loudspeaker shall be provided for monitoring receiver operation. Each loudspeaker shall be switchable to individual receivers and shall have a volume control which includes an off position.

3.5 Central and Local Processor Software. The software shall include executive, operational, maintenance and utility programs.

3.5.1 Executive Software. The executive software shall monitor, sequence, schedule and control SFBCS operation. The executive shall control interrupts, real time clock, error sensors, I/O devices and console interfaces.

3.5.2 Operational Software. The operational software for the receiver and transmitter groups shall provide for all operating requirements for burst communications. Operational software functions are outlined in

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paragraphs 3.4.8 of this specification.

3.5.3 Maintenance Software. Maintenance software shall be provided for detection and localization of system or equipment faults and, where possible, the recovery from those faults. Both the central and local processors will contain diagnostic or monitoring routines to be used in the maintenance of the Communication Central. These diagnostics shall allow for system level checks, sub-system level isolation, or, in some instances, module and board level isolation. BITE within equipments shall be utilized, if available. On-line and off-line maintenance routines, and manual procedures where needed, shall be provided. On-line maintenance routine shall be controlled by the executive software.

3.5.3.1 System On Line Software Maintenance Routines. As a minimum, on-line software maintenance routines shall include a self-check (BITE) sequence for the central and local processors. On-line routines shall monitor for fault conditions within other system components, where such interfaces are available. For those equipments with appropriate interfaces, and equipment status listing shall be available on the sytem plasma displays or computer terminals, by keyboard command. The operational status of the system transmitters shall be continuously monitored and displayed, as required in paragraph 3.4.7.2 (4). When a transmitter fault is detected, the scheduled outgoing message traffic shall automatically be switched to a back-up transmitter, in accordance with paragraph 3.4.7.2. The above functions shall be interwoven with the normal system functions and shall not disrupt regular message traffic or system operation.

3.5.3.2 Computer Subsystem Off-Line Software Maintenance Routines (GFP). The Government will provide (GFP) the contractor with Burst Communications Diagnostic System (BCDS) maintenance tapes to assist maintenance personel in troubleshooting and testing the AN/TSC-99 Computer subsystem, including the processors, I/O chassis, mass memory, magnetic tape unit, and line printers. This automated disgnostic tape is controlled via either the sytem plasma displays or computer terminal and uses displayed menus to prompt the operator through each diagnostic test. The BCDS provides complete testing and fault isolation to the board level, with a visual display or printout of the test results.

3.5.3.3 Bite (Built In Test Equipment) Available BITE in equipment shall be utilized to isolate defective modules with limited isolation to specific components on modules. This will be used either with the on-line routine, off-line routine, or for manual isolation consistent with the available processor interfaces within the various equipments.

3.5.4 Utility Software The utility software shall provide functions not

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required during normal operation. These functions shall include but not be limited to addition, deletion and modification of stored data via I/O devices and the dumping of selected areas of a memory to an output device.

3.5.5 Software Modification Lockouts shall be provided to prevent any software modifications by user personnel. The capability of performing on-site software adjustments shall be solely the prerogative of post deployment software support personnel. Strict configuration management principals shall be applied to system software over the communications central life cycle.

3.5.6 Software Documentation The software documentation requirements shall be in accordance with the contract notes and CDRL's, DD Form 1423.

3.6 General Configuration. The contractor shall plan the configuration of equipment in the shelters in accordance with the details of this specification. A drawing of the proposed layout for the transmitter group shall be submitted to the government for approval before starting any installation. A drawing of the proposed layout for the receiver group shall be submitted to the government which will be followed by a conference with the contractor. As the result of agreements reached in the conference on the proposed configuration, the contractor shall construct a mock-up of the receiver shelter to the extent necessary to demonstrate that the human factors are acceptable. The mock-up should have provision for installation of a cover to represent a ceiling in order to verify operator working space. Upon approval of the government of the final configuration, the contractor shall proceed with the installation.

3.6.1 Shelters. The bare shelter for the Transmitter Group shall be the S-280 B/G and will be supplied by the Government unless otherwise specified in contract. The Receiver Group shelter shall be provided by the contractor and shall be the S-280C/G as presently manufactured for the Government by Gichner Mobile Systems except for exterior length dimensions which shall be extended to 17 1/2 feet.

3.6.2 Shelter Facilities. The bare shelter shall be modified to include installation of all power and signal entrance boxes, power distribution, all power and signal wiring and cabling, racks, bracketry, tables, desks, lighting, provision for the installation of ECUs, transporter, and chemical protector, and all common items referenced in 3.13. This shall be the shelter facility, and it contains all items necessary to permit installation of operational equipments. Internal shelter structural members shall not be relocated or cut during the modification.

3.6.2.1 Rings for Cable Grips. A row of "D" rings shall be provided on the exterior of the shelter for attachment of cable grips used with the RF transmission lines to prevent strain on the antenna entrance panel

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connectors. A suitable mounting plate shall be provided for the rings in order to transfer the tension loads on the rings to the structural members of the shelter.

3.6.3 Entrance Panels and Cover Assemblies. Shelter entrance panels shall be recessed so that no portion of the connectors protrude from the plane of the shelter wall. A hinged cover shall be provided to protect the signal entrance panel from weather conditions during operating modes and enclose the signal entrance panel during transit. Exterior cover assemblies shall be constructed similar to the methods of SC-D-539681. The support to hold cover open is shown as item 6 of drawings SC-D-681010, sheet 2. A typical installation is contained in SC-D-539711.

3.6.3.1 Signal Entrance Box. The box, item 1 of SC-D-681010, shall be constructed similar to methods of SC-D-681011 and associated drawings. The switches, connectors, and cable facilities shown in SC-D-681010 do not apply to this communication central. The holes shown on the drawing for connectors, switches and the rectangular opening for receptacle U-187/G shall be omitted. The recess for switches is not required and may be omitted. The quantity of binding posts shall be as defined in paragraph 3.3.21 and 3.4.26. An appropriate opening in the box shall be provided for the passage of wires to the shelter signal wire duct.

3.6.3.1.1 The contractor shall add bracketry to the box as required for installation of surge protectors and RFI filters required by the paragraph 3.4.26. The box shall be wired in an appropriate manner for the installed components.

3.6.4 Wiring and Raceways. All interconnecting wiring for all components and accessories of the shelter shall be secured in cable raceways, with suitable clamping devices. Power and signal wiring shall be in separate raceways. Raceways shall be attached to the shelter with blind rivets to the shelter members only. All access areas shall be free of wiring.

3.6.5 Interconnection Cable and Wiring. The shelter facility shall include all cabling, cords, wiring, junction boxes, and connectors necessary to interconnect all internal components in their operating position. Cabling between components and shelter entry panels, including connectors, shall also be provided. All connectors exposed to the weather shall have captive waterproof covers, and in no case shall cables be concealed in the walls or floor of the shelter. All floor spaces within the shelter shall be kept free of cables. Sufficient cable slack shall be provided to rack-mounted equipments to permit the operation of equipment in the extended position. Cable strain relief shall be provided for all interconnecting cables. Internal wiring installation shall be in accordance with MIL-E-45782.

3.6.6 Equipment Racks. Shelf type equipment racks with each corner supported shall be provided. Use of cantilever mounting racks is unacceptable. Equipment should be capable of installation into racks with mounting inserts and spacing of MIL-STD-189. Racks shall be aluminum weldments. Rack attachment to the mounting members in the shelter panels shall be through shock or resilient mounts to intermediate Zee or hat

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members. Where feasible, tilt-lock, slide mounts should be used for mounting the equipment.

3.6.7 Mounting Inserts. All mounting inserts located in the structural members of the shelter, shall be installed without the use of any auxiliary materials such as adhesive and when installed the head shall not protrude more than .062 inch from the inside skin of the shelter. When installed each insert shall be capable of withstanding the tensile load and torque load test given in Section 4, Product Assurance. There shall be no damage to the insert threads, and the insert shall not have rotated from its original position, and there shall be no permanent distortion of the panel, mounting member, or insert bead or threads. Insert threads and outer face of insert head shall be free of paint. Inserts shall be selected from SC-D-539667, SC-D-681075, SC-D-681076, and DC-D-681080 for nut, blind, rivet. Inserts used in the floor, the door, or in any external surface of the shelter shall be closed and of stainless steel.

3.6.8 Seal, vapor. All joints and raw edges of the framing for the blower exhaust, power and signal entrance boxes, and air-conditioner cut-outs and any other framed openings shall be sealed to provide a vapor barrier against the entrance of moisture to the core material of the shelter panels and to the interior of the shelter. All joints and edges shall be assembled and sealed in a manner to prevent collection and retention of moisture. All exterior rivets, screws, and fasteners which penetrate the shelter skin shall be dipped in sealer prior to installation. The sealer shall be Silastic 738 as made by Dow Corning or equal using the manufacturer's recommended primer.

3.6.9 Weight. The weight of any shelter assemblage, fully equipped for transport, shall not exceed 8,000 pounds, exclusive of the weight of the lift and tie-down slings and the transporter/transporter interface hardware. The weight distribution of the transmit and receive shelters with all equipment installed shall not exceed a lateral imbalance greater than 5% and 22% respectively, as determined by the location of the center of gravity relative to the center line of the shelter, using the curbside as reference. The imbalance shall be calculated as follows:

$$\% \text{ Imbalance} = \frac{Dc - Dg}{Dc} \times 100$$

Where Dc = distance of centerline from the curbside wall.
Dg = distance of center of gravity from curbside wall.

The fore and aft weight imbalance shall not exceed 14% for the transmit shelter and 5% for the receive shelter, calculated as above except the distance shall be from the fore end wall of the shelter. The fore end of the shelter shall be the heavier end.

3.6.10 Lateral Stability. The center of gravity of the mobile assemblage shall be such that the angle between a vertical line (90 degrees to base line) through the center of gravity and a line drawn from the center of gravity to the nearest outer base ground contact point, shall be not less than 25 degrees.

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3.6.11 Transporter. Each shelter shall be equipped with a mobilizer having wheels and towing device for highway travel as well as jacks for leveling the shelter at the installation site. The recommended mobilizer is the model having the nomenclature: Dolly Set, Lift, Transportable Shelter, M-832. If no attachment exists, two towing rings shall be added on the rear to provide attachment points for hooking an aircraft winch cable for towing the assembly backwards up a ramp and into the aircraft.

3.6.12 Transport. Each fully equipped shelter mounted on an M-832 transporter shall be transportable by C-130 aircraft and towable by 5 ton truck. The shelter and transporter dimensions shall not exceed specifications as specified in 463L loading system and shall satisfy basic design considerations and cargo restraint considerations as specified in USAF Specification Bulletin 518a. Each fully loaded shelter mounted on an M-832 transporter shall satisfy Type III mobility as specified in MIL-M-8090.

3.6.13 Environmental Control. The cooling and heating requirements shall be met using dual air conditioners which may have heating capability together with additional electric heater units as necessary to meet the requirements of this specification. Military type, Type II, Size B, Class 1, 18,000 BTU horizontal, compact units shall be used which operate from 230 volt, 50 and 60 Hz, single phase, electric power.

The design and installation of the ECU system shall be submitted to the Government for approval. Operation of the system shall be accomplished without objectionable noise or drafts. Operation of the system shall be such that there shall be no detrimental effect upon the performance of other electrical apparatus installed in the shelter. The air conditioner units shall be externally mounted on the front wall of the shelter and shall be provided with flexible discharge hoses to direct the condensate to the ground. Waterproof insulated blank off panels of shelter wall type construction shall be provided for each of the two openings exposed by removal of the air conditioners when dismounted for maintenance or other reasons. A permanent storage position for the covers will be provided, and both the storage location and the covers will be suitably marked. The method of air conditioner installation shall be such as to permit convenient and simple interchange of the units. They shall be accessible for routine servicing, preventive maintenance, and adjustment of the outside fresh air damper, without removal from its installed position. Vibration isolation mounting shall be provided. Acoustical absorbent material and other means as required shall be utilized to minimize noise from the units (and supply air ducts) to meet the human engineering requirements of this specification.

3.6.13.1 ECU Transformer In order to provide the proper voltage for the operation of the air conditioners specified above, a power transformer shall be provided for each ECU to convert the 120 volt shelter voltage to 230 volts, 3 wire, service. The transformer shall be in a rigid RFI inclosure.

3.6.13.2 Shelter Cooling
The air conditioning units shall have the capacity to meet the

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following conditions:

(a) With all power off and external ambient temperature stabilized at 120°F for a period of 24 hours, the work space temperature shall be reduced from +120°F to +90°F within 2 hours after power to the ECU only is turned on.

(b) Maintain work space temperature not to exceed +90°F dry bulb with relative humidities in the range of 5% to 75% and an external ambient of +120°F. This requirement shall be met while all equipment normally used within the shelter is in operation as well as the presence of operating personnel plus the effects of solar radiation.

3.6.13.3 Equipment Cooling. Unit package components shall be arranged to permit proper cooling of all components with the cooling air flow through the equipment cabinet in a vertical direction. The pressure drop of the air through the individual package shall not exceed 0.3 inches of water when a sufficient volume of air is being moved through the equipment to properly cool the components. This requirement does not apply to transmitters which have integral air circulation facilities.

3.6.13.4 Heating. The heating capacity shall be capable of raising the temperature of all personnel workspace as follows:

-25°F to 0°F within 2 hrs
0°F to +32°F within 2 hrs. Equipment heat may be utilized.
+32°F to +65°F or +70°F within 3 hours. Equipment heat may be utilized.

With the equipment operating, the heating capacity shall be capable of maintaining dry bulb temperature within personnel workspace of 65°F or greater with an exterior ambient of -25°F. This requirement shall be met using electric heating.

3.6.13.5 Ambient Noise. With all equipment operating, including but not limited to the two air conditioners in high cool mode, rack cooling fans, paper tape punches, and line printers, the maximum allowable ambient noise shall not exceed the levels set forth for Category "D" of MIL-STD-1474. Under typical operating conditions, with both air conditioners in the high cool mode, all rack cooling fans on, one paper tape punch punching, and one line printer printing, the nominal ambient noise shall not exceed 82 db (A) (using A-weighted level meters).

3.6.13.6 Control Panel. The control panel (or panels) on the air conditioners shall contain a thermostat control and a selector switch for selecting the mode of operation. The controls shall be accessible from inside of the shelter. If necessary to improve accessibility, the control panel(s) may be separate from the units and located in another part of the shelter. If electric heater units are used which contain a thermostat they need not be controlled by the above thermostat.

3.6.13.7 Dust Control. All air distribution systems shall be provided with sufficient collectors, filters or other protective devices to remove

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all dust particles detrimental to operating personnel, which may be met under any condition of military operation. Traps and positive closures shall be provided at all exterior openings to afford protection from wind blown dust, snow, and water during operation or storage of the vehicle or shelter under wind conditions specified herein.

3.6.13.8 Chemical protection. Provision shall be made in the curbside wall of each shelter for connection to the air delivery duct of a Chemical Corps M6A1 collective protector. The connection fitting shall be in accordance with drawing ESD-209948 or the equivalent which will provide the same fit and function. The exterior cover shall be stenciled, "CHEM PROTECT PORT" or equivalent.

3.6.13.9 Air Tightness. The equipment shelter, both operating (summer operation) and non-operating, shall be capable of maintaining air to a static pressure of at least 0.5 inches of water while air is being brought in from the collective protector at the rate of 300 cfm with personnel fresh air ports and doors closed. The system shall be capable of meeting these requirements when in operation with the electronic equipment cooling air set to take air from outside the enclosure and exhausted to the outside. The system shall be capable of meeting these requirements within 5 minutes when connected to a Chemical Corps M6A1 Collective Protector.

3.6.13.10 Pressure Relief Valve. A pressure relief valve shall be provided in the shelter door to relieve ventilation air in any quantity up to 30 cubic feet per minute. If required by the valve, a cover shall be provided to close the valve when the shelter is not being ventilated. If required by pressure relief valves, instructions regarding operations for air shipment shall be stenciled on the outside.

3.7 Workmanship. New design and/or modifications to existing commercial or Government equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs.

a. In MIL-P-11268.

General requirements for plastic material and plastic parts.	Process for soldering
Wiring and cabling, including:	Cleaning.
Slack Protection	Riveting. General requirements for securing of parts.
Clearance	Cleaning of welding
Splicing	Process of welding
Connections, general	Mechanical operation of controls

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Grounding, general

Shielding on wire and cable

Flux and cleaning agents for soldering

b. In this specification:

Marking & labeling
Finish

Gears

Self-tapping screws

3.7.1 Welding. Welded joints shall be defined by the drawings and shall be such that grinding on the finished weld will be unnecessary except where grinding is specified on the drawings. Inert-gas-shielded arc welding (MIG) using helium, argon, or a mixture of the two shall be used for arc welding of aluminum. Welds shall conform to MIL-W-6858 Class B and MIL-W-45205 Class B as applicable; surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free of scale, paint, grease, and other foreign materials. Welds shall have through penetration and good fusion and shall be free of scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Inert-gas-shielded arc welding shall be used for welding of aluminum except where spot welds are used. Welded assemblies shall be cleaned to remove any scale, oxidation products, and excess flux. Any acid used in cleaning shall be completely neutralized and removed.

3.7.2 Thermal Coefficient of Expansion. Metals having different thermal coefficients of expansion shall not be used in inter-related assemblies where dimensional change through the service condition temperature range has a degrading effect on equipment operation or maintainability.

3.7.3 Ferrous Parts. The use of ferrous alloys shall comply with MIL-STD-454, Requirement 15, Ferrous Alloys, Corrosion Resistance.

3.7.4 Dissimilar Metals. The selection and protection of dissimilar metal combinations shall comply with MIL-STD-454, Requirement 16, Dissimilar Metals.

3.7.5 Marking & Labeling. The system and equipment marking and labeling shall conform with MIL-M-13231 and MIL-STD-1472, respectively, except for those commercial items which already have markings/labels. Front panel markings shall be Group I, as described in MIL-M-13231. Clear, concise, and durable markings/labels shall be provided for all system rack/shelf mounted equipments, controls, lamps, switches, fuses, jacks, connectors, test points, and other components in accordance with the paragraphs titled, LABELING, of MIL-STD-1472. All design and integration areas that present a potential safety hazard to personnel shall be clearly and concisely marked/labeled in accordance with MIL-STD-454, Requirement 1, and safety requirements of paragraph 3.18 herein. Equipment within the system shall be serial numbered and each equipment rack in a shelter shall be numbered. Each shelter of the AN/TSC-99 shall have its transmitter or

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receiver group nomenclature, but, carry the AN/TSC-99 system serial number.

3.7.5.1 Visibility. Whenever practicable parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.7.5.2 Instructions. Suitable instructions shall be supplied within each shelter to cover the following information.

a. Preparation of assemblage for operational use.

b. Preparation of the assemblage for transport.

3.7.5.3 Aircraft Loading data plate. Each fully equipped shelter without transporter shall be provided with data plates imprinted with the dimensions which locate the centers-of-gravity of the shelter facility and the assemblage in the three reference planes; the gross weight of the shelter facility and assemblage; and the bearing pressure on the bottom of the skids. There shall be a second data plate providing the same information for the combined shelter and transporter.

3.7.6 Finish. Finishes shall be in accordance with MIL-F-14072.

3.7.6.1 Exterior Surfaces. The exterior surfaces of facilities and/or assemblages such as covers of boxes, side flaps on covers, and shrouds for air-conditioners and blowers, will have the same finish as the exterior surfaces of the bare shelter (Finish P510.1V per Spec MIL-F-14072).

3.7.6.2 Exterior Hardware. Exterior hardware for facilities and/or assemblages such as entrance boxes, latches, braces, underside of covers, etc., will be finished the same as exterior surfaces. These components will be painted prior to assembly to the shelter and any touch up will be with the lusterless green enamel of the shelter.

3.7.6.3 Interior Items.

a. Bracketry. - Semi-gloss light gray enamel, per FED-STD 595, No. 26440. If the bracketry is ferrous material Finish P213.5 per Spec MIL-F-14072 or if it is aluminum Finish P513.2 per Spec MIL-F-14072.

b. Storage Boxes. - Finishes shall be in accordance with MIL-F-14072.

(1) Aluminum Alloys

(a) Apply steps 1 and 2 of P513.2 to interior and exterior.

(b) Apply steps 3 and 4 of P513.2 to exterior only.

3.8 Blackout Curtain. The shelter shall contain a fireproof blackout curtain of a type that will not billow, mounted on the inside of the doorway.

3.8.1 Viewport. The doors in both shelters shall contain a one-way wide angle viewport (view from inside to outside). Provision shall be made to prevent the viewport from being covered by the door ventilation hatch cover when in open position.

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3.9 Door Bell or Buzzer. A suitable audible notification system (located inside the shelter) with an activation assembly located outside the shelter by the door shall be installed in the receiver group and transmitter group shelter. This requirement may be met by an electrically operated bell or buzzer or a mechanically operated bell.

3.10 Storage Space. Enclosed storage facilities consisting of partitioned drawers and cabinets shall be provided for operating spares, technical manuals, spare parts, miscellaneous hardware, patch cords, and spare printed circuit cards. An enclosed storage facility shall be provided at the operator's position for storage of materials required by the operator such as instructions, paper, pencils, etc. These facilities shall not consist of transit cases, but shall be part of the integrated design of both transmitter and receiver shelters.

3.10.1 Program Cassette Storage. The transmitter and receiver shelters shall be equipped with an enclosed storage space for holding the necessary number of magnetic tape cassettes which shall be supplied with the communication central for processor programming in accordance with paragraph 3.4.8.2.1. The storage compartment shall be easily accessible and at such a level so as to permit easy identification of needed cassettes.

3.10.1.1 Space for additional cassettes shall be provided in the event that the same model is used for digital recording of incoming burst message traffic.

3.10.2 Message Tape Storage. Space shall be provided for storing a supply of paper tape which is used for the teletype tape punch and other message tape punches in the receiver group.

3.10.3 Page Printer Paper Storage. Space shall be provided for storing a supply of paper for use with the page printers installed in both the receiver and transmitter shelters.

3.10.4 Storage Facilities at COMSEC Position. Enclosed storage (for daily supply) for crypto pads and for reels of crypto tape used by the COMSEC operator shall be provided.

3.10.5 Message Cassettes. Space shall be provided for the storing of spare magnetic tape cassettes used for recording manual CW traffic.

3.11 Chairs. Operators' chairs shall be provided in each shelter in the quantities specified below. Provision shall be made for securing each chair to the shelter during transit. The preferred type of chair for the receiver group is the rotary without arms but with a non-tilting seat and with a tilting adjustable back similar to Type II, Class 3, Style B of Federal Specification AA-C-00275E with a padded type seat. A long wearing fabric is preferred for the seat covering rather than a plastic material such as vinyl. An acceptable type of chair for the transmitter group is a folding chair with a padded seat which can be readily stowed.

4 ea for Receiver Group
2 ea for Transmitter Group

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3.12 Test Equipment. One Multimeter AN/USM-223 or equivalent (GFP) shall be provided in the transmitter shelter and one in the receiver shelter. The contractor shall provide recommendation for added standard or non-standard test equipment. If the manufacturer of a complex item such as the transmitter has a portable test set available as a catalog item for on-site diagnostic test and rapid fault isolation, it should be included in the recommendations.

3.13 Miscellaneous Items. The following ancillary items shall be provided for each shelter. The items shall be mounted in accessible locations following the human engineering principles of MIL-STD-1472/13. Those items having an asterisk shall be assembled in one area as a safety group. Other items may be located in the same or other areas if more convenient.

Quantity	Item
2 each	Ground Strap (heavy braid, 1 inch wide, 10 feet long)
1 each	Hammer, hand, sledge, blacksmith, double face 6 lb.
1 each	Clock, nominally 6 1/2 inch diameter (approximately) face, sweep second hand, non-electric type.
1 each	Light, extension w/guard, w/25 foot cord and 50 W lamp.
1 each	*Lantern, wall, 6 volt, Justrite Mfg. Co., Mod. 2106-7, or equal.
1 each	Sharpener, pencil, mechanical cutting type.
1 each	*Axe, pick head, 6 lb, 5" cutting edge.
4 each	Clothes hooks
1 each	Brush, hand, dusting
1 each	Hand Vacuum Cleaner
2 each	Basket, waste paper, metal type
1 each	Extinguisher, Fire, Charged, 5 lb, w/safety lock, Walter Kidde Co. #5T2 or equal
1 each	Bracket, Fire Extinguisher, Walter Kidde Co. Model RB-1
3 each	Ash tray, unbreakable, table type.
1 each	*Kit, First Aid, General Purpose, Davis Emergency Co., #458-0-24, or equal.
2 each	Ground Rod, Hubbard Co. #9416, or equal.
1 each	Broom, Osborn Mfg. Co. #1820, or equal.
1 each	Rack for 4 rifles in receiver shelter.
1 each	Rack for 2 rifles in transmitter shelter.

3.13.1 Bulletin Boards. A bulletin board for posting of memos using thumbtacks shall be provided on the roadside wall in the receiver shelter. The height and width may be controlled by the available space. One bulletin board shall be provided in the transmitter shelter.

3.14 Primary Power. Each shelter shall be provided with a complete electrical system, which shall conform to the following:

a. National Electrical Code (NEC) NFPA No. 70-1978

b. MIL-E-4158E. (This military specification shall govern in

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case of conflict with any other document).

c. The primary power AC subsystem shall provide power for all the electronic circuitry, and charging current for batteries. The power for accessories, such as the air conditioner, heater, exhaust blowers, lights, outlets, and other features incorporated within the shelter, shall be provided by the primary subsystem. Adequate protection shall be provided to prevent damage to or malfunction of the equipment due to voltage or current surges, over/under voltage, or external radio frequency (RF) sources. (See 3.17.1 for RFI filtering.)

3.15 Primary Power, Shelters. Each shelter shall be designed for a nominal 120/208 volts AC, three-phase, seven conductor, 50 and 60 Hertz shelter input power. The 4th, 5th and 6th conductors shall be a neutral and the 7th a ground. The neutrals and ground are tied together only at the generator or prime power transformer. Loads of all phases, except transmitter loads which are intermittent, shall be balanced within 10 percent. The equipment shall operate satisfactorily when utilizing appropriate engine-generators having characteristics listed in MIL-STD-1332 for Type 1, Classes 1 and 2, Modes I and III generator sets. In addition, the equipment shall operate satisfactorily when furnished three-phase, five-wire power from commercial sources with the following steady-state ranges: 120/208 volts AC ± 10 percent, 47 to 63 Hertz. Other worldwide single and three phase voltages shall be accommodated through use of transformers.

3.15.1 Three Phase Power Transformers. In order to operate worldwide from commercial sources, a tapped three phase, 50-60 Hz, Y connected, transformer having adequate KVA rating to power the transmitter shelter and one to power the receiver shelter shall be provided for each AN/TSC-99. Transformers shall be located outside the shelter and be suitable for outdoor environment. Stud type terminals shall be provided for five wire input and output connections. Each transformer shall be dolly mounted having wheels with one swivel pair suitable for moving along a paved roadway by manpower. Taps shall be provided for the following phase to phase line input voltages: 190, 220, 260, 380, and volts. Output shall be three phase Y-connected 208 volts AC.

3.15.2 Single Phase Power Transformers. In order to operate worldwide from commercial sources where three phase power is not available, a tapped single phase, 50-60 Hz transformer having adequate KVA rating to power the transmitter shelter and one having adequate KVA rating to power the receiver shelter shall be provided each AN/TSC-99. Transformers shall be located outside the shelter and be suitable for outdoor environment. Stud type terminals shall be provided for input and output connections. The phases in the power cable from the shelter will be connected as single phase at the transformer. Each transformer shall be dolly mounted having wheels with one swivel pair suitable for moving along a hard surface roadway by manpower. Taps shall be provided for the following line voltages: 110, 130, 150, 200, 240, 400, and 500 volts. Output shall be single phase 120 volts AC.

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3.15.3 Single Phase Operation. All installed equipment including ECU's and transmitters shall be single phase with loads distributed among the phases. For operation from a single phase power source, it shall be possible to connect all three phases of the shelter power cable to one terminal and all the neutral conductors to the other terminal of the single phase transformer of paragraph 3.15.2 or to a commercial source to form one single phase load. This provision is intended to permit connection to a foreign commercial power system where three phase service is not available.

3.15.4 Power Entrance Panel. A power type AN connector, 120/208 volts AC, 3 phase, seven conductor, 50/60 Hertz, shall be provided on the shelter exterior with a captive protective metal cap. The power entry receptacles shall be located on the outside, door-end, road side of the shelter; shall be accessible when the shelter is mounted on vehicular transporters; and shall have labels or markings providing the following information:

a. Input or output
b. Pin arrangement
c. Voltage and frequency rating

3.15.5 Shelter Power Cable. One seven conductor power cable 100 feet long shall be provided on a reel for each shelter of the communication central. One end shall have a connector to mate with the shelter receptacle. The other end shall have screw clamp, solderless lugs.

3.15.6 Ground Terminal. A ground terminal with a pressure type connection accommodate size 1/0 conductor shall be located on the power entrance panel on the exterior of the shelter.

3.15.7 Shelter Power Distribution. Primary power in each shelter shall be distributed so that the power to any major equipment or group of equipments can be turned off and removed without affecting power to the rest of the equipment in the shelter. The system shall include a power entrance connector panel, a power distribution panel(s), and all hardware, wires, and cables to provide a complete distribution system. The distribution box shall include, but not be limited to, a main circuit breaker to remove all power except air conditioners to the shelter and individual circuit breakers for interrupting power to subfeeder systems or individual equipment as necessary. The distribution box shall have safety provisions to prevent accidental contact with exposed "live" terminals when the access panel is opened. All inter- connecting wires and cables shall be mounted in raceways, conduit, and other appropriate devices.

Warning labels shall be applied to the power distribution box and the ECU transformer box to prevent safety hazards during maintenance as follows:

Above the ammeter on the power distribution box:
CURRENT MEASUREMENTS DO NOT INCLUDE AIR CONDITIONER OR HEATER CURRENT.

On access panel to the ECU transformer box:

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ECU TRANSFORMER CIRCUIT BREAKER BEHIND PANEL.

CAUTION: REMOVE POWER FROM SHELTER BEFORE ENTRY FOR MAINTENANCE.

3.15.7.1 Distribution Circuits. Separate power distribution circuits in separate raceways shall be provided as part of the shelter as follows:

a. Shelter entrance receptacle to distribution box.
b. 120 Volts AC 50/60 Hertz, single phase
c. Any other circuits as determined by the contractor.

3.15.7.2 Main Circuit Breaker. The main circuit breaker shall have an interrupt capacity of at least 15,000 amperes at 50 or 60 Hertz. This requirement also applies to the ECU transformer circuit breakers if they are not on line with the main circuit breaker.

3.15.7.3 Distribution Box. The panel shall be a complete distribution center, with automatic and manual controls, instruments and protective devices. It shall include, but not be limited to, the following:

a. It shall be located and configured so as to permit convenient operation of all manual controls and convenient observation of all instruments.

b. Manual reset circuit breakers shall be provided for all circuits. Fuses shall not be permitted. Multipole circuit breakers shall be of the common trip type having a single operating handle.

(1) A main circuit breaker shall be provided as identified in 3.15.7.2.

(2) Separate branch circuit breakers shall be provided as required for interior convenience receptacles, equipment as necessary,lighting, heaters, etc.

c. A voltmeter, and ammeter, each switchable from phase to phase, shall be provided to monitor input power. These instruments shall be ruggedized and sealed. The switch shall be provided with an "OFF" position. The scale increments and ranges of both meters shall be chosen to permit accurate readings.

3.15.8 Lights. Lighting circuits, with necessary fixtures, switches, lamps, and filters shall be provided as a part of all shelters. Internal lighting shall provide suitable illumination for operation and maintenance in accordance with the provisions of MIL-STD-1472A. Emergency lighting of at least 50 watts per shelters shall be provided. The lighting should be designed to avoid direct illumination of video display units.

All shelter lighting circuits shall be provided with door interlocks that shall shut off the main shelter lights and turn on a blue low-level light when the shelter door is opened. The interlock switch shall be

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capable of being overridden by an auxiliary switch for continuous lighting during nonblackout conditions. Lighting fixtures shall be located to minimize personnel contact and be provided with guards where necessary to prevent bump/laceration hazards.

3.15.9 Convenience Outlets. At least four readily accessible duplex (straight three-prong type with ground) convenience outlets shall be provided inside each shelter. The wiring shall be sized to provide 20-ampere continuous service. Two separately protected circuits shall provide power for these outlets.

3.15.10 Equipment Power Cable Plugs. The power cords for all equipment drawing less than 15 amperes shall use the straight 3 prong (one ground) plug on all AC power cords.

3.15.11 Battery Power. A battery power system, floating on a charger shall be provided which will automatically keep certain equipment energized as specified in paragraphs 3.3.15 (transmitter group), 3.4.15 (receiver group), 3.15.8 (lights) when primary AC power fails. The type battery selected by the contractor shall be submitted to the Contracting Officer for approval. The battery compartment shall comply with the safety provisions of paragraph 3.18.1.4

3.15.11.1 Shelter Battery Chargers. Battery chargers with adequate electrical protection shall be provided as an integral portion of both the receiver shelters and transmit shelters. The battery charger shall be capable of fully charging fully discharged batteries within 24 hours without damaging the batteries. Protection shall be provided to ensure that the batteries are neither overcharged nor damaged. The battery charger shall not present a power factor load to the AC power source of less than 0.75 lagging. Automatic provision for cyclic charge and recharge shall be provided if required by the batteries selected.

3.15.12 Exterior Power Receptacles. Receptacles shall be provided on the exterior of each shelter for connection of the cable used to supply 115 volt power to the antenna tower warning lights. Provision shall be made to connect two cables to each shelter.

3.16 Service Conditions. It is recognized that the communication central will be an assemblage of commercial equipments. However, the communication central will be deployed worldwide so that the design objective for selection of hardware, air conditioners, and heaters should meet the range of conditions given in the following paragraphs.

3.16.1 Low Temperature. Both internal and external shelter ambients are given in case any equipments use external air for cooling.

3.16.1.1 Storage and transit: -50 degrees F.

3.16.1.2 Heaters turn-on: -50 degrees F.

3.16.1.3 Equipment power turn-on: 0°F shelter interior ambient with shelter exterior ambient of -25°F.

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3.16.1.4 Equipment fully operational: +32°F shelter interior ambient with shelter exterior ambient of -25°F.

3.16.2 High Temperature. Both internal and external shelter ambients are given in case some equipments use external air for cooling.

3.16.2.1 Storage and transit: +155°F.

3.16.2.2 Air Conditioner turn-on: +120°F shelter interior ambient with +120°F shelter exterior ambient.

3.16.2.3 Equipment fully operational: +90°F shelter interior ambient with shelter exterior ambient of +120°F.

3.16.3 Solar Radiation. Equipment and air conditioners fully operational: +120°F shelter exterior ambient with solar radiation of 360 Btu/ft^2/hr.

3.16.4 Relative Humidity. The equipment shall be fully operational without use of the air conditioners at 95% to 100% relative humidity at an ambient of 100°F.

3.16.5 Fungus. Materials which are not nutrients for fungus shall be used wherever possible such as those given in MIL-STD-454B, Requirement 4. If it is necessary to use nutrient materials, these materials shall be treated by a method that will render the resulting exposed surface fungus resistant and pass the fungus test given in MIL-STD-810.

3.16.6 Altitude. Atmospheric pressures corresponding to the following limiting conditions:

a. Storage and transit: sea level to 40,000 feet.

b. Operating: sea level to high ground elevation of 10,000 feet.

3.16.7 Transportation. The transmit and receive shelters, with installed equipment, shall be capable of withstanding the following mechanical environment:

a. Vibration: All types of roads, cross country terrain, water, and fixed wing and rotor aircraft. In addition, equipment subassemblies shall be free of damaging resonances at frequencies below 55 Hz.

b. Shock: The equipment shall withstand the shock associated with vehicular and air transportation (including loading and unloading). Individual subassemblies shall withstand shock associated with bench handling.

c. Air Transport: The AN/TSC-99 shall survive without damage the air transport conditions of 3g fore and aft horizontal acceleration, 2g vertical upwards acceleration, and 4.5g vertical downward acceleration. The quantity 1g is equal to the weight of shelters.

3.17 Electromagnetic compatibility

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3.17.1 Shelters. Shelters shall provide at least 60 dB of attenuation to electric and magnetic fields and to plane waves over the frequency range of 0.15 to 10,000 MHz with the doors closed but all other covers open. Modification kit, MK-1079/K, DL-SC-A-621051, shall be installed to enhance the shielding characteristics of the basic shelter. All openings cut into the shelter walls or roof, such as exhaust ports, ventilation ports, air conditioner intake and outlet ports, etc., shall be provided with a waveguide beyond cutoff honeycomb filter to maintain the 60 dB shielding integrity of the shelter structure. A multiple section RFI filter shall be installed on each phase and neutral of the AC input power leads. The filters shall be installed on the inside of the shelter at the shelter power input panel.

3.17.2 Bonds and Grounds. All contractor furnished equipment shall be designed with adequate provisions for bonding the units to the shelter structure without adversely affecting maintainability or interchangeability. Bonding shall not be accomplished through screws connecting the equipments to mounting racks. Bonding impedances measured across any two adjacent metal surfaces shall be 2.5 milliohms or less. Shock mounted units or those employing vibration isolators shall have a low impedance bond, 2.5 milliohms or less, between the unit and equipment ground plane. Bonding jumpers shall be of the solid metal type and as short as possible, however, in no case shall the length to width ratio of the jumper be in excess of 5 to 1. Surfaces being bonded together shall be prepared and cleaned by removing all anodic film, dirt, grease, paint or other high resistance materials from the immediate area to insure negligible radio frequency (RF) impedance between adjacent metal parts. Upon completion of the bonding assembly and ascertainment of the 2.5 milliohm bonding impedance, the completed assembly shall be refinished with its original finish or other suitable protective finish in accordance with MIL-F-14072.

3.17.3 Cables and connectors. Case and chassis mounted connectors shall have an approved RF conductive finish and shall be installed such that the bonding impedance between connector shell and case or chassis shall not exceed 2.5 milliohms. Connectors which support cables which interface with other subsystems not supplied under this development description shall utilize pin filters to alleviate or minimize any undesirable emissions on the interface lines. Connectors, which terminate overall cable shields, shall be provided with an EMI backshell for peripheral bonding of the cable shield.

3.17.4 Intra-system electromagnetic compatibility. Each equipment as installed in the shelters shall not cause undesirable response, degraded performance or malfunction of any other equipment within the shelter when the system is fully operational and all equipments are performing their intended functions. The equipment shall be exercised through all modes of operation as required by this specification.

3.17.5 Electromagnetic interference. The communication central shelters with all equipments installed shall comply with the following emission and susceptibility requirements of MIL-STD-461A, Notice 4:

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(g,h)CE02	CS02	(d)RE02	(f) RS03
	(b)CS03	(d,j)RE02.1	
(h,i)CE04	(c)CS04	(e)RE03	
(a,h)CE06	CS06		

a. Transmitters in the Key-Down mode shall have spurious and harmonic emission levels as specified in 3.3.2.

b. Intermodulation levels shall be as specified in 3.4.2.

c. Spurious response rejection levels and image frequency rejection shall be as specified in 3.4.2

d. The transmit shelter shall be tested on the door and rear ends for radiated emissions at a test distance of 3 meters. Emissions shall be measured with the antenna alternately positioned for horizontally and vertically polarized emissions when linearly polarized antennas are utilized above 30 MHz. The air conditioner and heaters shall be on and cycling during the test.

e. Method RE03 may be used in lieu of CE06 for transmitter spurious and harmonic outputs.

f. Equipment inside shelter shall not malfunction, exhibit degraded performance or undesirable response when the shelter is immersed in electric fields of the intensity specified below. The door end of each shelter, with the shelter door open, shall be immersed in the field and both horizontally and vertically polarized fields shall be utilized above 30 MHz if linearly polarized antennas are employed.

Shelter	Frequency	Field Intensity	Modulation
Receive	14 kHz to 12.4 GHz	5 Volts/meter	None
Transmit	14 kHz to 400 miles	5 Volts/meter	None

g. Conducted emissions shall not exceed requirements by more than 22dB.

h. Transmit shelter test only.

i. Conducted emissions shall not exceed requirements by more than 22dB at 64 kHz and below, wideband.

j. RE02.1 tests shall be conducted from 14 kHz to 10 GHz.

3.18 Safety engineering. The system and equipment design criteria shall be in accordance with Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, "Occupational Safety and Health Standards"; Requirement 1 and 8, MIL-STD-454; and paragraphs entitled "Labeling" and "Hazards and Safety" of MIL-STD-1472, and shall include but not be limited to the following requirements. Those requirements apply to both commercial equipment and new design modifications.

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3.18.1 Personnel safety. The system shall be designed to minimize the probability and severity of injury to personnel throughout the systems life cycle. Personnel safety is an absolute requirement of the equipment and shall not be considered a candidate for trade-off.

3.18.1.1 Mechanical safety. The system design shall include the safety provisions called for in paragraph entitled "Mechanical" of Requirement 1, and 8, MIL-STD-454 and the following:

a. Exposed edges and corners shall be rounded.

b. Adequate provisions for lifting and handling shall be provided in accordance with paragraph entitled "Unit Design for Efficient Handling", MIL-STD-1472.

c. Adequate safeguards shall be installed to prevent inadvertent entrapment of body parts and clothing in moving parts.

d. Color coding of indicator lights shall conform with Table II, MIL-STD-1472. This requirement does not apply to a manufacturer's standard catalog equipment.

3.18.1.2 Electrical safety. The system design shall include the safety provisions called for in paragraph entitled "Electrical" of Requirement 1 and 8, MIL-STD-454, and the following:

a. Accidental contact with current-carrying metal components, terminals, and like devices (greater than 70 volts) shall be controlled by the following:

(1) Wire connections (bare terminals) shall be protected with insulated sleeving (MIL-I-23053/5).

(2) All contacts, terminals, and like devices other than wire connections shall be provided with barriers or guards to prevent operating and maintenance personnel from accidental contact. These barriers or guards shall be manufactured from a transparent, nonconductive material (insulation resistance shall be not less than 1000 megohms) and shall be marked with the approximate highest voltage (nearest round number) and current type which may be encountered upon removal. The marking shall be in a format in accordance with American National Standards Institute ANSI Z35.1-1972 and worded as follows: "CAUTION (Maximum Voltage applicable) Volts AC" (or DC when applicable).

(3) Small openings shall be provided (where needed) in guards or barriers for maintenance testing without the removal of the guards. The voltage encountered through these openings shall be marked on the guard or barrier.

b. Components over 500 volts shall be competely enclosed and interlocked with no bypass devices permitted and shall be marked in accordance with paragraph entitled, "Warning Markings", Requirement 1 and 8, MIL-STD-454.

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c. Socket connectors, designed to prevent personnel from accidental contact with potentially hazardous voltage, shall be used on electrically energized contacts such as power sources.

d. When radio interference filters are used in the power input, filters shall contain only capacitors between each side of the line and chassis which are sufficiently low in value to allow less than 5 milli-amperes of current to flow to the chassis under the most adverse conditions of maximum power frequency and maximum voltage permitted by the individual test equipment specification.

e. Positive acting, automatically actuated discharging devices shall be provided to discharge high voltage circuits and capacitors to 30 volts within 2 seconds or less.

3.18.1.3 Grounding. The system design shall comply with the following provisions to assure all noncurrent-carrying metal parts are grounded by non current-carrying conductors and color coding conforms to MIL-STD-1857:

a. Requirement 1, MIL-STD-454, paragraphs entitled as follows: "Ground Potential", "Grounding", "Grounding to Chassis", "Shielding", and "Grounding rods".

b. All power input or interconnecting cables shall be provided with a green insulated equipment ground wire (EGW) used only for grounding noncurrent-carrying metal parts of equipment. Test equipment leads shall have provisions to insure that the chassis of equipment under test is also at ground potential.

c. The equipment ground wire shall be terminated at both ends in the same manner as the other conductors and shall be of the same or larger size and voltage rating as the largest of the other conductors or the sum of all paralleled conductors. It shall carry no current at any time other than during a ground fault.

d. All grounding terminals shall be green.

e. All power return wires (neutral) shall be white and white shall be used only for power return wires (neutrals).

f. The equipment ground wire shall not be connected to the grounded power return (neutral) at any point other than the generator or the secondary of the power distribution transformer.

g. Color coding for all power cables shall comply with the following:

(1) Single phase a.c. service (such as 115 volt):

Black (hot)
White (neutral)
Green (equipment ground wire)

NOTE: Ground is essential; 3-wire cable required.

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(2) Three phase a.c. circuits:

Black (hot) Phase A
Red (hot) Phase B
Blue (hot) Phase C
White (neutral)
Green (equipment ground wire)

NOTE: Ground is essential; 5-wire cable is required.

(3) Single phase, tapped a.c. (such as 110/220):

Black (hot)
Red (hot)
White (neutral)
Green (equipment ground)

NOTE: Ground wire is essential; 4-wire cable required.

3.18.1.4 Chemical safety. The system design shall include the following provisions:

a. Battery enclosures shall be of such construction that in the event of battery venting, electrolyte will not be expelled in either liquid or droplet form.

b. If a battery is utilized which may emanate explosive gasses (i.e., hydrogen), the battery box design will be such that it insures no leakage of the gas into the main equipment or into other sources of ignition. Battery box venting will be provided to minimize the explosive gas concentration in the battery box.

c. No material or component which during any phase of the life cycle will evolve toxic gasses, vapors, fumes, or droplets shall be used in the system.

d. No treating materials shall be used that are identified in the Toxic Substance List (published by National Institute for Occupational Safety and Health) that will produce toxic effects via the respiratory tract, eye, skin (cutaneous and subcutaneous), or mouth. This includes handling the treated material during fabrication, transportation, operation or maintenance of the parts in which the treated materials are used or during use of the finished parts when used for the purposed intended.

e. Battery Compartment Venting. Batteries that produce toxic or explosive gasses shall be vented to the outside of the shelter through a vent port. A fan shall be provided to insure adequate exhausting of battery compartment gasses, but the rate of air flow must not result in negative pressure within the shelter under the conditions given in paragraph 3.6.13.9, Air Tightness. The exhaust vent to the outside shall be of such a design as to prevent water from entering the shelter during storage or transport. If a hinged

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cover is used it shall be equipped with an interlock and an indicator to prevent equipment operation with the cover closed.

3.18.1.5 Microwave and X-radiation safety. The system shall be designed to include the criteria in TB MED 270 and the following:

a. Personnel exposure to X-radiation shall be limited to 2.0 mrem/hr at 5 centimeters (cm) from the radiating source.

b. Personnel exposure to microwave radiation shall be limited to 10 mw/cm^2.

3.18.1.6 Radioactive material restriction in optical products. Optical products shall contain no thorium or other source materials as defined by Title 10, Code of Federal Regulations, Part 40, in excess of 0.05 percent by weight (500 ppm) or any other added radioactive materials. Optical product are defined as optical glass constituents or raw materials, optical glass, optical glass components such as windows, filters, reflector, prisms, beam splitters, lense elements and fiber optics, optical assemblies, and optical coatings except for IR objective lenses. Radioactive material is defined as radioactive material per item in excess of the concentrations listed in Schedule A, Title 10, Code of Federal Regulations, Part 30, or in quantities greater than 0.001 microcurie.

3.18.2 Equipment Safety.

a. The system shall be designed to minimize equipment damage, degradation of efficiency or mission failure due to the following conditions:

(1) Operator induced errors.

(2) Improper cabling.

(3) Power failure or electrical overstress on components.

(4) Secondary failure.

(5) Installation, storage, operation, handling, maintenance, and transportation.

b. Specific design techniques for equipment protection shall include the following:

(1) Electrical overload protection shall be in accordance with Requirement 8, MIL-STD-454. All fuse positions shall be marked with the rated current capacity of the fuse to be employed therein. Fuse positions for delayed-action fuses shall have the additional designation "SLOW". For "bayonet" type fuses, the contact nearest the insertion end shall be at ground potential when the fuse is removed.

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(2) Contacts used on chassis, panels, or cable entrance connectors shall be recessed to prevent breakage or damage. Separate connectors to be used for AC/DC input/output when operated on both.

(3) Positive means shall be provided to prevent the inadvertent reversing or mismating of fittings, couplings, fuel, oil, hydraulic, and pneumatic lines; mechanical linkage; and instrument leads and electrical connections.

3.18.3 Environmental Safety.

a. Personnel shall be provided an acoustical environment which will not cause personnel injury, or cause fatigue, or in any other way degrade overall system effectiveness.

b. Personnel exposure to noise shall be limited in accordance with MIL-STD-1474 (MI). See Paragraph 3.6.13.5

c. The fact that a component which contributes to the overall noise may be government furnished equipment shall not eliminate the requirements set forth in MIL-STD-1474 (MI).

d. Personnel exposure to vibration shall be limited in accordance with paragraph entitled "Vibration" or MIL-STD-1472.

e. No radioactive material shall be utilized without prior expressed approval of the Contracting Officer (potential sources of radiation include luminous dials/markings, electron tubes, surge arrestors and lenses).

f. Hazardous environments - the equipment shall be specially designed in accordance with Chapter 5 of the National Fire Code, Volume 6 to preclude accidental ignition or explosion when a probability exists or operation in a Class 1 (gasses or vapors), Class II (combustible dusts), and/or Class III (ignitable fibers and flyings) hazardous atmosphere.

3.18.4 Interlock. Interlocks shall be provided to prevent access to the interior of the equipment, or components thereof, without removal of all voltages exceeding 130 volts between any two points of the equipment or between any point and ground. Interlocks shall conform to paragraph titled "Interlocks" in the Switches Section of Requirement 1, MIL-STD-454.

3.18.5 Warning Labels. During maintenance where voltages exceed 70 volts, protective devices and warning labels shall be provided for in accordance with MIL-STD-454. Warning labels outlining use of grounding sticks at maintenance access points of each HF power amplifier shall be provided.

3.18.6 Grounding Sticks. Grounding sticks in accordance with paragraph entitled "Shorting Rods" in the Electrical Section of Requirements 1,

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MIL-STD-454, except that transparent sleeving shall be optional, shall be provided and stored in such a manner that they shall be readily available for use to discharge capacitors in the transmitter high voltage supply.

3.18.7 Rubber Mat. Each shelter shall be equipped with a removable heavy duty rubber mat that covers the open floor. The purpose of this mat is to insulate maintenance personnel from the shelter metallic structure.

3.19 Maintenance

3.19.1 Maintenance Storage. The contractor shall provide recommendations for storing the following items for transportation purposes: receiving and transmitting antennas, RF transmission lines for antennas, power cables and switch boxes, reels of wire, shade tarpaulins and hardware, spare parts and replacement equipments, air conditioners, test and repair equipments and fixtures, spare parts and other miscellaneous items.

3.19.2 Maintenance. The base station shall be designed to provide for corrective maintenance actions based upon repair by replacement of parts/ modules/components, on-line adjustments, or a combination of both at organizational level. The design shall be such that major corrective maintenance actions will not take place on a malfunctioned item before replacement is switched or patched on-line. Preventive, or scheduled maintenance, may be performed on redundant units or standby units, within the specified safety criteria. Under no circumstances shall scheduled maintenance actions remove the base station from an operational status. The base station design, or its associated documentation, shall provide for efficient methods and techniques of fault determination, isolation, correction or repair, and repair verification; readiness assurance tests and inspection; servicing (replenishment of consumables after operation); and scheduled maintenance at all maintenance levels.

3.20 Reliability - The contractor shall assure that the AN/TSC-99 reliability characteristics are fully considered during system design and integration and the best achievable utilizing the latest commercial state-of-the-art. During subsystem (subcontractor) equipment selection, the contractor shall assure that the equipment RAM characteristics are considered as a major source selection factor.

3.21 Reliability Program. The contractor shall conduct an approved reliability program in accordance with MIL-STD-785. The reliability program shall be responsive to the requirement set forth in preceding paragraph. Reliability program plan, analysis (including mathematical model) shall be in accordance with the contract and Statement of Work of DD Form 1423.

3.21.1 Subcontractor and Supplier Reliability Programs. The contractor shall be responsible for all subcontractor and supplier reliability programs to the extent described in MIL-STD-785. Reliability requirements shall be included in subcontractor and supplier specifications.

3.21.2 Commercial Equipment Reliability. The contractor shall be responsible for assessing the reliability of all assemblies and equipment

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purchased as commercial items and constructed according to best commercial practice. The contractor shall gather appropriate reliability assurance data from the commercial vendors or in lieu thereof, the contractor shall derive these data. In no case will the contractor construe his overall system reliability requirements as excluding the reliability of commercial equipment.

3.22 Maintainability

3.22.1 Maintainability Requirements.

(1) The Mean Time to Repair (MTTR) at organizational level shall not exceed 30 minutes. Repairs should be limited to locating and replacing faulty components, i.e., cable and cord assemblies, accessory items, and other replacement pluck-out items and piece parts external to equipment.

(2) The MTTR at Direct Support/General Support level shall not exceed 60 and 120 minutes respectively. Repairs shall be limited to locating and replacing faulty modules and pluck-out items internal to the equipment. Troubleshooting and repairs shall be limited to replacement of mechanical piece parts such as switches, relays, etc., and circuit cards.

3.22.2 Maintainability Program. The contractor shall develop and implement a Maintainability Program in accordance with MIL-STD-470. The maintainability program shall be responsive to the specific maintainability requirements set forth in preceding paragraphs. Maintainability program plan, analysis and predictions shall be in accordance with contract and Statement of Work, DD Form 1423.

3.22.3 Subcontractor and Vendor Maintainability Programs. The contractor shall be responsible for all subcontractor and vendor maintainability programs to the extent described in MIL-STD-470. Maintainability requirements shall be included in subcontractor and vendor specifications.

3.22.4 Commercial Equipments. The contractor shall be responsible for the maintainability of all assemblies and equipment purchased as commercial and constructed according to best commercial practice. The contractor shall gather appropriate maintainability assurance data from the commercial vendor or in lieu thereof the contractor shall derive these data. In no case shall the contractor construe his overall system maintainability requirement as excluding the maintainability of commercial equipments.

3.23 TEMPEST Design Requirements. Receiver Group OR-218/TSC-99 shall meet the BLACK Line Conduction Requirements and Limits of NACSEM 5100, Compromising Laboratory Test Standard; Electromagnetics (title UNCLASSIFIED; document CONFIDENTIAL). For design and testing purposes, the RED signals are defined as any classified plain-text signals which are processed by the COMSEC PC boards and KW-7/TSEC components of the off-line COMSEC equipment and secure teletypewriter.

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4.0 QUALITY ASSURANCE PROVISIONS

The inspection of the AN/TSC-99 Communications Central shall include tests and evaluations from the time of unit manufacture to the final on-site acceptance of each system. The system, as referred to herein, includes the Receiver Group, the Transmitter Group, miscellaneous hardware, and software.

The contractor's inspection program shall be designed to verify the electrical, operational, mechanical, and environmental characteristics of each unit, subsystem, and complete system, on both an individual and integrated basis. The purpose of the inspection program is to assure that each system, when installed at its site, complies with or exceeds the requirements specified herein and in the contract. Testing of units, subsystems, and the complete system is required for determining performance, effectiveness, compatibility of mated units, operational integrity in specified environments, and operational characteristics of each. Inspection includes in-plant and on-site tests and evaluations of hardware and software. Unit, subsystem, and system testing shall be performed in-plant and installation shake-down and final acceptance testing shall be performed on-site.

4.1 Responsibility for Inspection. Unless otherwise specified in the RFP or contract, the contractor shall be responsible for the performance of all inspection requirements as specified herein, as well as those requirements which may be determined after submission of the detailed test plans by the contractor.

Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification or included in the approved test procedures where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

This does not relieve the contractor of his responsibility for performing any additional inspection necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspections to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.1.1 Supplier Quality Control. The contractor shall be responsible for insuring that all items supplied, whether his own or subcontracted items, meet the performance requirements of this specification.

4.2 Classification of Inspection.

a. First Article Inspection (4.3).

b. Quality Conformance Inspection (4.4)

c. Packaging Inspection (4.15)

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4.3 First Article. The first article inspection shall be conducted by the contractor on the first completed communications central, prior to acceptance testing of any other system produced.

4.3.1 First Article Inspection. The first article inspection shall consist of the inspections specified in Table I and shall be performed in an order satisfactory to the Government.

4.3.2 First Article Data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

TABLE I. FIRST ARTICLE INSPECTION

	Inspection	Req. Para.	Test Para.
1.	Burn-In (Subsystem Equip)	--	4.4.1
2.	Visual and Mechanical	3.6, 3.7,3.14 3.15, 3.18,	4.4.2, 4.4.2.1, 4.4.2.2
3.	Subsystem Equipment Tests	3.3, 3.4	4.4.3
4.	System Tests (In-Plant)	3.3, 3.4, 3.5	4.4.4
	Base Station		4.4.4.1
	-Base Station - Outstation		4.4.4.2
	-Satellite System	3.4.5	4.4.4.3
	-Teletype System	3.4.17, 3.4.18	4.4.4.4
5.	Software Testing (FQT)	3.5	4.5
6.	Temperature Testing	3.6.13, 3.16.1, 3.16.2, 3.16.3	4.6
7.	EMI Testing	3.17	4.7
8.	TEMPEST Testing	3.23	4.8
9.	Air Tightness	3.6.13.9	4.9
10.	Ambient Noise	3.6.13.5	4.13
11.	Safety	3.18	4.14
12.	Transportability	3.16.7	4.10
13.	On-Site Contractor Installation and Demonstration	3.3, 3.4, 3.5	4.11
14.	Packaging Inspection	5.0, 6.2	4.15

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4.4 Quality Conformance Inspection. The contractor shall assure that each Communications Central produced is inspected for conformance to all the inspections specified in 4.4.1 through 4.4.4.4. This does not relieve the contractor of his responsibility for performing any additional inspections which are necessary to control the quality of the product and assure compliance with all specification requirements.

4.4.1 Burn-In (Subsystem Equip.). The contractor shall assure that each piece of subsystem equipment, containing active (electrical/electronic) circuitry, is subjected to a minimum 48 hours on-time burn-in period. Burn-in may be performed by sub-contractors provided sufficient data is obtained to satisfy the data requirements of 4.4.1.3. During the last 24 hours of burn-in, the equipment must operate failure free. Up until this time the equipment will be allowed to accumulate failures. Each equipment which fails during the final 24-hour period shall be repaired and returned to test until it successfully survives a 24-hour period without failures. Failures which occur during burn-in shall be analyzed and recorded. For commercial equipment, on which burn-in has previously been performed, burn-in may be waived following Government approval of the alternate burn-in procedure. This requirement does not apply to government furnished equipment.

4.4.1.1 Burn-In Conditions.

a. Burn-in shall be performed at a constant ambient temperature of $35^{o} \pm 5^{o}C$ ($95^{o} \pm 10^{o}F$).

b. Equipment on-off cycling shall be performed throughout the burn-in period, as follows: Turn equipment on and let temperature stabilize; hold for 3 hours, then turn equipment off, and again let temperature stabilize. Repeat this cycle throughout burn-in period.

c. During equipment on-time, the equipment shall be operated in a mode which simulates typical mission operation. If more than one mode is possible, modes shall be switched for equal on-time for each mode, over the total burn-in period.

4.4.1.2 Burn-In Operational Tests. An abbreviated operational test shall be performed on the equipment prior to burn-in, prior to and after the last 24 hour failure-free period, and on a daily basis, during burn-in. The operational test should check those parameters necessary to assure correct equipment operation.

4.4.1.3 Burn-In Data. Burn-in data shall be submitted as required in the contract or purchase order.

4.4.2 Visual and Mechanical Inspection. Each subsystem equipment and completed Communications Central shall be visually and mechanically inspected for those defects listed in MIL-STD-252 and specified herein.

Parts, components and complete equipments shall be inspected for workmanship (see 3.7), mechanical construction, loose nuts and screws, application of specified treatments and miscellaneous defects. The

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equipment and controls shall be inspected for proper mechanical and electrical operation. Wiring, soldered connections, grounds (see 3.18.1.3), welds (see 3.7.1), finishes (see 3.7.6), etc. shall be inspected for workmanship and conformance to the safety requirements of 3.18. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with 3.7.5 Examination shall also be made for other visual of mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall be classified by the Government as major or minor in accordance with the definitions of MIL-STD-105.

4.4.2.1 Weight Distribution. During first article testing only, the fully equipped shelter assemblies shall be tested for compliance with paragraphs 3.6.9. The imbalance shall be measured as follows:

a. Weigh the shelters, without the mobilizers, using Type A, Load-O-Meter road scale or equivalent, placed under the skids near each end of the shelter. Measure the distance of the load cell locations from the curbside and the fore end of the shelter.

b. Record weight readings for each cell.

c. Determine the location of the center of gravity in the horizontal plane from both the curbside and fore end using the following:

$$Dg = \frac{W_1d_1 + W_2d_2 + W_3d_3 + W_4d_4}{Wt}$$

Where Wn = Weight reading of individual load cell.
Wt = Total of all load cell readings.
d_n = Load cell distance from curbside wall (or fore end) corresponding to Wn.

d. Determine lateral and longitudinal balance in accordance with paragraph 3.6.9, Weight.

4.4.2.1.1 The center of gravity for the assemblage shall be determined for both with and without the mobilizer as required for the aircraft loading data, paragraph 3.7.5.3 and lateral stability, paragraph 3.6.10. The contractor shall provide the procedure for determining the center of gravity in the first article test plan.

4.4.2.2 Mounting inserts. During first article testing only, four of each size insert mounted in opposite panels, e.g., floor and ceiling or side and side shall be tested with the axial load, specified in Table II, applied to eye bolt in the insert. After axial load is removed the specified torque load (Table II) shall be applied to a bolt in the insert. (Note: Bolt shall lock on threads in insert and not "bottom" nor shall the bolt head be tight against head of insert.) After test, the requirements of 3.6.7 shall be met.

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TABLE II

First Article Insert Loads

Insert Size	Tension (lbs)	Torque (in lbs)
5/16	2000	40
1/4	1300	23

4.4.3 Subsystem Equipment Testing. Each subsystem equipment produced or procured as part of the Communications Central, shall be electrically tested to assure proper operation as compared with the equipment's technical description, literature, or specification(s). The testing may be performed either by the contractor or sub-contractor (see 4.1.1). Prior test data may be used, if available, for commercial off-the-shelf equipments which have a proven history of operation. Where suitable prior test data (test results) are not available, or where the test data are determined by the Government to be inadequate, the contractor shall perform additional testing, as needed. Test data shall indicate principally the operational capability of the equipment to perform its stated function. This requirement does not apply to Government furnished equipments.

4.4.4 System Tests (In-Plant). Following integration of the receive and transmit shelter assemblies, the contractor shall perform system tests to verify proper system operation and compliance with the system requirement. specified in Section 3. The system tests shall also verify the inter-operation of the receive shelter group, the transmit shelter group, and the outstation equipment. Outstation equipment (GFP) shall consist of the AN/PRC-70 and AN/PRC-74 transceivers, Radio Set AN/PSC-3 and the Digital Message Device Group (DMDG), OA-8990( )/P. The following tests shall be performed as a minimum.

4.4.4.1 Base Station Loop-Back Test. During this test, the transmitters shall be terminated in dummy loads, the receivers shall be connected through multicouplers to antenna entrance panel, and the transmit group shelter and receive group shelter shall be coupled together through an attenuator for HF communications. In this configuration, system testing shall be performed to verify all aspects of receiver shelter/transmit shelter interoperability.

4.4.4.1.1 Standard Test Conditions.

(1) Test conducted at room temperature and shelters located near each other.

(2) The intershelter message and control lines shall simulate 5 miles of telephone wire.

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(3) Three individual dummy loads shall be connected to antenna entrance panel outside the transmitter shelter to dissipate transmitter power.

(4) A long wire to the antenna entrance panel of the receiver shelter may be used to pick up radio signals at safe levels from the transmitter group dummy loads by stray coupling. Direct coupling to the dummy load through an attenuator may be used.

(5) Adequate electrical power shall be supplied to the transmitter shelter to operate two transmitters at full power simultaneously with the third on standby.

4.4.4.1.2 Standard Test Message Tapes (without encoding).

Note: The processor must send each message on a tape twice.

(1) Tape A shall consist of 10 messages, each having seven five-letter groups plus an individual five-digit identification number.

(2) Tape B shall consist of 4 messages, each having 200 five-letter groups including one individual five-digit identification number.

4.4.4.1.3 CPU Control - Manual. For compliance with requirements, demonstrate the following control functions using the CPU keyboard:

(1) Receiver frequency - Set each receiver to 3 different frequencies (one receiver at a time).

(2) Line and antenna switch operation including:

a. Transmitter group message line switch

b. Transmitter group antenna switch matrix

(3) Transmitter frequency - Set each transmitter to 3 different frequencies (one transmitter at a time).

(4) Transmitter power - Set each transmitter to high and low different power output levels (one transmitter at a time). Check power output holding CW key closed.

4.4.4.1.4 CPU Message Handling. The system shall be tested by inserting a message tape in a reader, entering release time and frequency instructions via the keyboard, sending the message automatically over the wire line to a transmitter or punched paper tape, and receiving the message back through a receiver to a tape punch. Performance shall be determined by verifying the received data for correctness.

The messages may be sent as rapidly as equipment permits. The following tests shall be performed as a minimum:

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Test 1: Use 1 transmitter and one receiver simulating scheduled operation.

a. Use 10 different radio frequencies (in a non-sequential pattern) over the frequency range. Schedule operating transmission times for both receiver and transmitter. Message release time may be as close as convenient. Send each of the messages on a different frequency.

b. Disable one of the pair of receivers in one diversity receiving terminal. Make sure that the receiver in use is connected to the test antenna.

c. Send message Tape A and verify correctness of the received message.

Test 2: Repeat test #1 but use only the second of the pair of receivers in the diversity terminal.

Test 3: Repeat test #1 but use both receivers simulating diversity operation. Both receivers may be connected to the same multicoupler.

Test 4: Use 1 transmitter and one receiver simulating scanning operation.

a. Use 10 different transmitter frequencies (in a non-sequential pattern) over the frequency range. Program receiver for scan mode and to lock on 7 of the 10 frequencies. Send each of the messages on a different frequency. Message release times may be as closed as convenient.

b. Use receivers in the second of two diversity receiving terminals and disable one of the pair. Make sure the receiver in use is connected to the test antenna.

c. Send message Tape A and verify correctness of the received message.

Test 5: Repeat test #4 but use only the second of the pair of receivers in the diversity terminal.

Test 6: Repeat test #4 but use both receivers simulating diversity operation. Both receivers may be connected to the same multicoupler.

Test 7: Use 2 transmitters and 2 receiving terminals for 1 tape.

a. Use one group of 10 frequencies for one transmitter-receiver link and a second group of 10 for the other transmitter-receiver link. Send the same message over both links simultaneously. Schedule transmission times for both transmitter and receiver. Verify correctness of the received messages.

Test 8: Repeat Test 7 but use the transmitter which was previously idle as one of the two in use.

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Test 9: Repeat Test 8 but use message Tape B and only a group of 4 frequencies for each link.

Test 10: Use 1 transmitter and 1 receiving terminal for fault demonstration. Use a group of 4 frequencies on a schedule and message Tape B.

a. During message transmission, disable the transmitter in use by turning off the plate voltage or by other convenient means. Determine if the message traffic is automatically switched to another transmitter. Verify that the message is correctly received two times.

Test 11: Use 1 transmitter and 1 receiving terminal and send messages from the transmitter shelter.

a. Use Tape A and 10 frequencies on a scheduled basis. Use transmitter shelter message facilities to enter the message Tape A and establishing a transmission schedule. Verify the correctness of the received messages.

b. Repeat test but use the second of the three transmitters.

c. Repeat test but use the third of the three transmitters.

Test 12: Demonstrate line amplifier adjustment range.

a. Connect one of the message circuits between the two shelters with a pair of wires having minimum length. Adjust line amplifier in transmitter shelter to the proper output level (for the transmitters) using a test signal from in the receive shelter.

b. Adjust the line amplifier in the transmitter shelter to the proper output level for that line simulating 5 miles of WD-1/TT wire using the same test signal.

c. Determine if the adjustment range of the line amplifier exceeds that required to obtain the levels required by procedures a and b above.

Test 13: Demonstrate the diagnostic procedures and printout for the transmitter group. The procedure shall be performed in the transmitter shelter and also on the CPU in the receiver shelter.

4.4.4.1.5 Voice, Manual CW, Tape CW. Demonstrate operation of the voice, manual CW, and punched tape CW, using base station loop-back configuration.

4.4.4.2 Base Station-Outstation Interoperability. During this test, an

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AN/PRC-70 and AN/PRC-74 transceiver (outstation) shall be coupled to the receive and transmit shelters (base station). Attenuators and dummy loads shall be used, as appropriate. An OA-8990 ( )/P, DMDG shall be used with each manpack radio to facilitate burst communications. All base station-outstation interoperation requirements shall be verified in this configuration. The capability of the base station to process 12 one-minute outstation message transmissions per hour, shall be demonstrated. The ability of the outstation to successfully receive base station messages shall be demonstrated by the following test:

Test 14: Demonstrate Base Station CPU message compatibility with outstation DMDG, OA-8990/P.

a. Couple the outstation equipment to the transmit shelter (base station).

b. With the outstation in the receive mode, send Tape A messages using a single transmitter. Verify the correctness of the received messages.

c. Repeat procedure b using Tape B. Compare the readout of the DMDG with the original tape, but disregard the overflow in the DMDG.

d. Couple the outstation equipment to the receiver shelter. Using the DMDG, send a message similar to Tape A and another similar to Tape B. Verify the correctness of the received messages.

4.4.4.3 Satellite Circuit Test. The object of this test is to verify the functioning of the satellite radio circuits in the receiver group without using the satellite radios which may not be available.

a. Connect the DMDG OA-08890/P SAT send and receive circuits to the satellite data circuits at the patch panel in the receiver group.

b. Send Tape A through the outgoing message system for satellite communication in the receiver group to the DMDG. Verify the correctness of the received message but disregard any part lost in the overflow in the DMDG.

c. Input Tape A messages into the DMDG but do not overflow. Send through the incoming message system for satellite communications in the receiver group. Verify the correctness of the received messages.

4.4.4.4 Teletype System Test. Demonstrate the secure teletype system by a loopback operation where the sending circuit is temporarily connected to the receiving circuit at the signal entrance panel of the receive shelter. Set up the two TSEC/KW-7 units in accordance with their instructions or use simulators, load a message tape into the tape reader, and send to the punch over the loop. Compare the received tape with the original for errors.

4.5 Software Testing. The contractor shall perform the following software tests/demonstrations in order to verify compliance with the

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software program requirements specified in paragraph 3.5 and applicable contract provisions.

4.5.1 Preliminary Qualification Test (PQT). A preliminary qualification test (PQT) shall be performed on each software module. The contractor shall inform the Government in writing of all changes to the modules and C level specification, after the PQT is performed. The purpose of the PQT is to provide a mechanism to both test and integrate the computer program components (CPC's) of the base station. Each successive PQT shall test the new CPC as it is integrated with previously tested CPC's. This approach shall provide for a controlled growth of the system software from a single CPC to a total integrated software system which is ready for Final Qualification Testing.

a. The contractor shall develop and submit a PQT plan. This plan shall define the schedule of the development, coding, testing, and integration of all CPC's.

b. The contractor shall develop and submit PQT procedures in accordance with the schedule set forth in the approved PQT plan. The procedures shall be submitted and reviewed concurrent with the review of the draft C-5 Specification for each CPCI. An approved PQT procedure is required before an "authority to proceed" release is granted to code the CPC.

4.5.2 Final Qualification Test (FQT). The purpose of the final qualification test (FQT) is to verify that individual programs function in accordance with the requirements of the Type B-5 specification and operator's manual, and to formally qualify the overall system computer operator's manual, and to formally qualify the overall system computer (processor) program. The FQT shall be performed during first article testing (FAT) (4.3) and the test results will be utilized during the first article accept/reject decision. The FQT/FAT shall be run using the AN/TSC-99 system and its associated peripheral equipment. The contractor shall verify those objectives of the FQT allowable by the in-plant environment, using simulated and/or actual inputs. Where practical, the contractor may combine parts of the FQT and the system tests of 4.4.4.

a. The contractor shall prepare and submit an FQT plan prior to the scheduled start of the FQT/FAT. The test plan shall provide for verification of the Type B-5 specification(s).

b. Based on the approved FQT test plan, the contractor shall prepare and submit FQT test procedures governing the conduct of the FQT.

c. The FQT/FAT shall be observed by Government representatives, unless otherwise notified.

4.6 Temperature Testing. The contractor shall assure that the complete communications central will operate within specification limits, at the temperatures specified in paragraph 3.16, and that the Environmental Control Units (ECU's) also function as required in paragraph 3.6.13. An abbreviated systems test shall be performed before, during, and after each of the following tests.

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4.6.1 High Temperature. The Communications Central shall be tested in accordance with Method 501.1, Procedure I of MIL-STD-810, with the following exceptions:

a. In Steps 1 through 4, the shelter doors, vents, etc. shall be in the open position.

b. In Step 2, the chamber temperature shall be maintained at +155°F.

c. In Step 4, the chamber temperature shall be lowered to +120°F. After temperature stabilization is reached, the shelter doors, vents, etc. shall be closed, and the shelter ECU's turned on and set to maximum cooling. The chamber temperature shall be maintained at +120°F.

d. In Step 5, the time required for the shelter interior to reach +90°F shall be noted for compliance with requirements. When the interior reaches +90°F, all the equipment within the shelter shall be turned on. After one additional hour with all equipment on and the chamber held at +120°F, the work space temperature shall be measured for compliance not to exceed +90°F. In addition, each equipment item shall be checked to determine that it is operational.

4.6.2 Low Temperature. The Communications Central shall be tested in accordance with Method 502.1-1, Procedure I of MIL-STD-810, with the following exceptions:

a. In Steps 1 through 4, the shelter doors, vents, etc. shall be in the open position.

b. In Step 2, the chamber temperature shall be maintained at -50°F.

c. In Step 4, the chamber temperature shall be raised to -25°F. After temperature stabilization is reached, the shelter doors, vents, etc. shall be closed and the shelter heaters turned on to start heating the shelter interior. The chamber temperature shall be maintained at -25°F. The capability of the shelter heaters shall be evaluated by assuring that the time to heat the shelter interior from -25°F to 0°F, 0°F to +32°F is less than two hours in each range and +32°F to +65°F is less than three hours. During the heating cycle, all the system equipment shall be turned on at 0°F and an abbreviated system test shall be performed at +32°F (Step 5).

4.7 Electromagnetic Compatibility Tests.

4.7.1 Shielding Effectiveness. The receive and transmit shelters shall be tested for compliance with the requirements of 3.17.1, using the approved measurement technique of MIL-STD-285.

4.7.2 Bonds and grounds. Prior to performance of EMI and intra-system compatibility testing, compliance with the bonding requirements of 3.17.2 and 3.17.3 shall be ascertained. Bonding impedance measurements shall be made with a Shallcross Meter, Model 670A, or equal and recorded for inclusion in the EMI/EMC test report. As a minimum, the following bonding

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impedance measurements shall be made:

a. Mounting rack to shelter structure

b. Equipment case or chassis to mounting rack or shelter structure

c. Power line filters-case to shelter

d. All shelter mounted connectors to shelter wall

4.7.3 Intrasystem Compatibility Test. Each shelter shall be tested to verify compliance with the requirements of 3.17.4. Tests shall be performed in accordance with a contractor prepared, government approved intrasystem electromagnetic compatibility test plan. This test may, however, be performed as part of the base station system tests, specified in paragraph 4.4.4.

4.7.4 Electromagnetic Interference. The receive and transmit shelters shall be tested for compliance with the requirements of 3.17.5. Test procedures shall be in accordance with the approved measurement techniques of MIL-STD-462, Notice 3, as implemented by a contractor prepared, government approved test plan.

4.8. TEMPEST Test. TEMPEST testing shall be performed by the contractor in accordance with the procedures of NACSEM 5100 and the contractor-prepared, government-approved TEMPEST Test Plan, submitted in accordance with Contract Requirements List DD Form 1423.

4.9 Air Tightness Test. Connect a duct from a blower to the port on the shelter where the chemical protector would be connected together with instrumentation to measure air flow and pressure. Close the fresh air intakes on the environmental control units. Close the door ventilator cover and plug the drain hole in the shelter floor. Open signal entrance panel cover and those on the transmitter shelter which admit and discharge cooling air to the outside. Turn on ECU blowers and battery box exhaust fan, if provided. Operate the air system at 300 cubic feet per minute plus or minus 10% into the shelter and note that a static pressure of at least 0.5 inches of water is maintained for 5 minutes.

4.10 Transportability.

4.10.1 Mobility Test. Demonstrate mobility and stability of the shelter assembled to its mobilizer by towing the assembly in a parking lot or other suitable area. Make at least 4 left and right turns at relatively slow speeds to simulate movement in an airport loading and storage area. Perform the maneuver of backing into a parking place at least 2 times.

4.10.2 Jacking Test. Lower one end of the shelter to the ground and jack the other end of the shelter to the maximum height using the mobilizer jack. Lower and repeat the lift reversing ends. Shelter and contents shall not show any signs of stress.

4.10.3 Transportability. This test shall be the shipment of the Communication Central AN/TSC-99 from the contractor's plant to the site

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where the contractor is required to install and demonstrate First Article and during Production Acceptance Testing (On-Site Contractor Installation Demonstration). The shelters shall be transported by air bag trailer. Upon arrival at the installation site the shelters and contents shall be examined for defects such as loose or damaged equipment and loose hardware.

4.10.4 Air Transport. A demonstration of air transportability shall be performed on the first article if the government makes a C-130 airplane available in the vicinity of the first article installation site. The AN/TSC-99 assembled on mobilizers shall be loaded and off loaded without requiring disassembly.

4.11 On-Site Contractor Installation and Demonstration

4.11.1 System Installation. Following the successful completion of all other first article tests and Government conditional first article approval, the contractor shall assemble the AN/TSC-99 in its normal shipment configuration and transport the system in accordance with paragraphs 4.10.3 to user test site (Identified elsewhere in the contract). Upon arrival at the test site, the contractor shall examine the receive and transmit shelters for compliance with 3.16.7, 4.4.2, and 4.10.3 and install the system in accordance with the user's test procedures and guidance.

4.11.2 System Demonstration. Following system installation, the contractor shall perform an abbreviated system demonstration, consisting of tests similar to those performed during the in-plant system test (4.4.4). The on-site tests, however, may include actual base station-outstation communications, instead of using loop-back techniques. The on-site system demonstration is the last first article test to be performed, is a deciding factor for first article approval, and therefore, must be witnessed by the Government.

4.12 Reliability and Maintainability Analysis. The contractor shall verify that the reliability and maintainability (RAM) requirements of paragraphs 3.20, 3.21 and 3.22, respectively, have been met by performing the RAM analyses, predictions, etc. in accordance with the contract provisions and statements of work of DD Form 1423.

4.13 Ambient Noise. With equipment operating as specified in 3.6.13.5, noise measurements shall be taken at each operator position, in each shelter, using an A-weighted level meter, to verify compliance with paragraph 3.6.13.5 herein.

4.14 Safety. The final system configuration and/or individual equipments shall be inspected to verify compliance with the requirements of 3.18 herein. The Government may elect to perform an independent safety inspection as a part of the First Article Testing.

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4.15 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging

b. Quality Conformance Inspection of Packaging

4.15.1 First Article Inspection of Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.15.2 Quality Conformance Inspection of Packaging.

4.15.2.1 Materials Inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.15.2.2. Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.15.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table III, "PACKAGING INSPECTION PROVISIONS". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality level of 4.0 percent defective.

TABLE III. PACKAGING INSPECTION PROVISIONS.

NO.	CHARACTERISTICS	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with spec.	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

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5.0 PACKAGING The requirements for packaging shall be in accordance with SPI 1G00415.

6.0 Notes

6.1 Intended use The equipment covered by this specifiation will be used world wide for military radio communications.

6.2 Ordering Data. Procurement documents shall specify the following:

a. Level A or B preservation and packing (See 5.0).

b. When first article rough handling tests are not required.

c. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packaging.

6.3 Level B Preservation. When level B preservation is specified, this level of protection will only be used under known faorable conditions during transportation, storage, and handling.

6.4 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.5 Government furnished property The following items are furnished in the quantity required by the procurement:

2 per Comm Central	Whip Antenna (3.3.1, 3.3.19)
2 per Comm Central	Multimeter AN/USM-223 or equivalent (3.3.1, 3.4.1, 3.12)
2 per Comm Central	Telephone TA-312/PT (3.3.1, 3.4.1, 3.4.19)
1 per Comm Central	Telephone TA-938/G (3.4.1, 3.4.19)
1 per Comm Central	Telegraph Terminal TH-22/TG (3.4.1, 3.4.18)
1 per Comm Central	Shelter S-280B (3.6.1)

6.6 Government loaned property The following items are loaned for the duration of the first procurement for use during the development and testing phases of the contract:

1 ea Radio Set AN/PRC-70 (4.4.4)
1 ea Radio Set AN/PRC-74 (4.4.4)
1 ea Digital Message Device Group OA-8990/P (4.4.4)
1 ea Coder KW-7/TSEC (Equipment: CONFIDENTIAL) (3.4.17)
1 ea Signal Converter KWX-11/TSEC (Equipment:CONFIDENTIAL)(3.4.17)
1 ea Digital Data Display Set AN/FYQ-90(V)1 (3.4.16)
NOTE: This item is unclassified when the NSA Processing Boards are removed. In the future, the item should be procured by the contractor.
1 ea Reader/Punch RD-486/U (3.4.16)
NOTE: In the future, this item should be procured by the contractor.

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1 ea Spectrum Monitor RSS-4 (3.4.1, 3.4.6)
2 sets Multicouplers p/o Antenna Group OE-316/TSC-99 (3.4.1,3.4.4.1)
1 ea Radio Set AN/VSC-7 (3.4.1, 3.4.5)

6.7 Equipment note The first procurement Communication Central shall be shipped without the following items which will be user installed.

Radio Set AN/VSC-7	(Not available for first time procurement)
Multicouplers(p/o OE-316)	(Users will transfer item from other systems)
Spectrum Monitor	(Users will transfer item from other systems)
NSA Processing Board	(User will requisition this classified item)

6.8 Antenna Note Except for the whip antenna, antennnas will not be included in the shipment of each communication central from the contractor. The Antenna Groups OE-316/TSC-99 and OE-317/TSC-99 will be provided by the government for the testing of the first article at the government test site (4.11).

6.9 Activity Symbols

CUSTODIAN:

Army - CR

PREPARING ACTIVITY:

Army - CR

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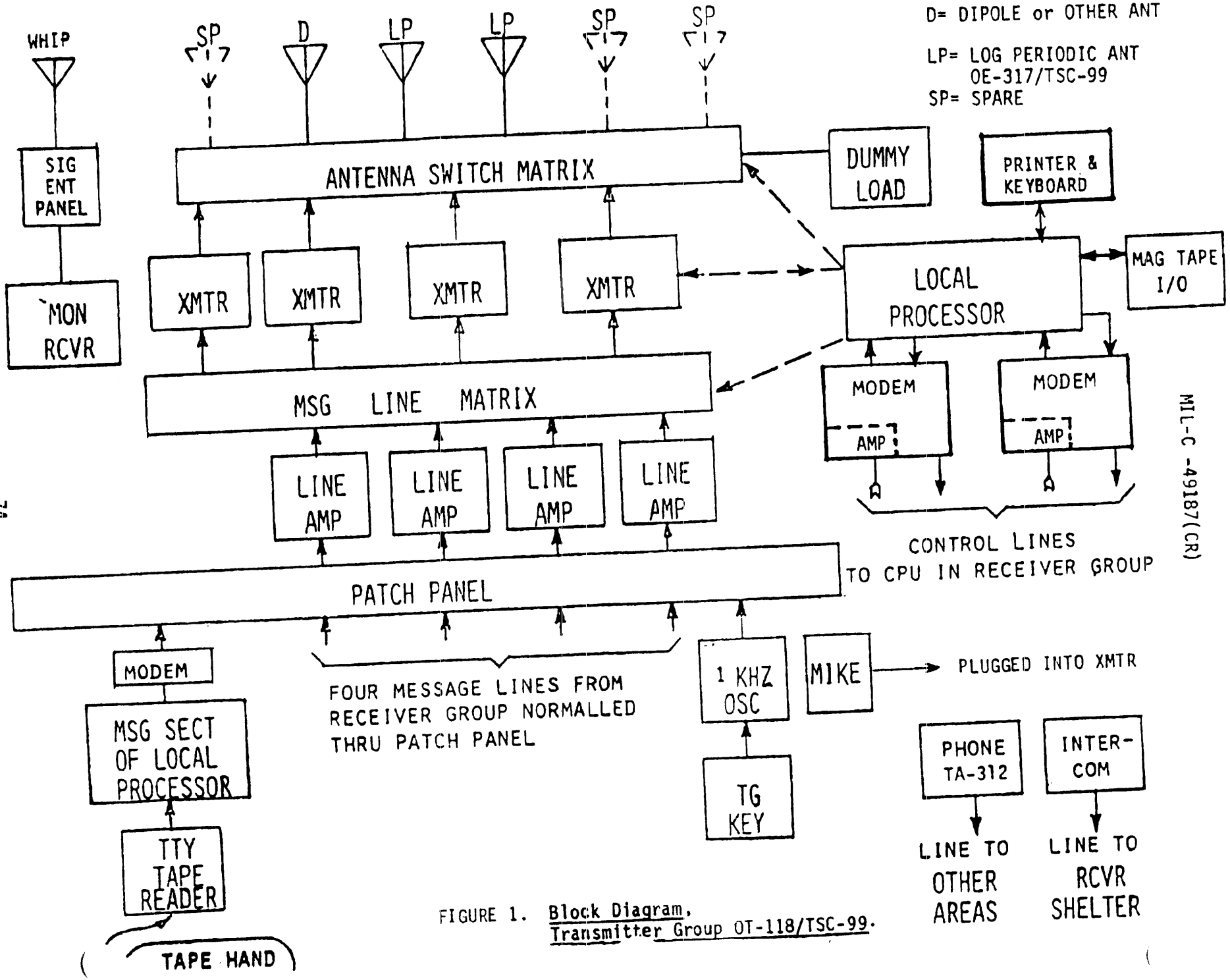


FIGURE 1. Block Diagram, Transmitter Group OT-118/TSC-99.

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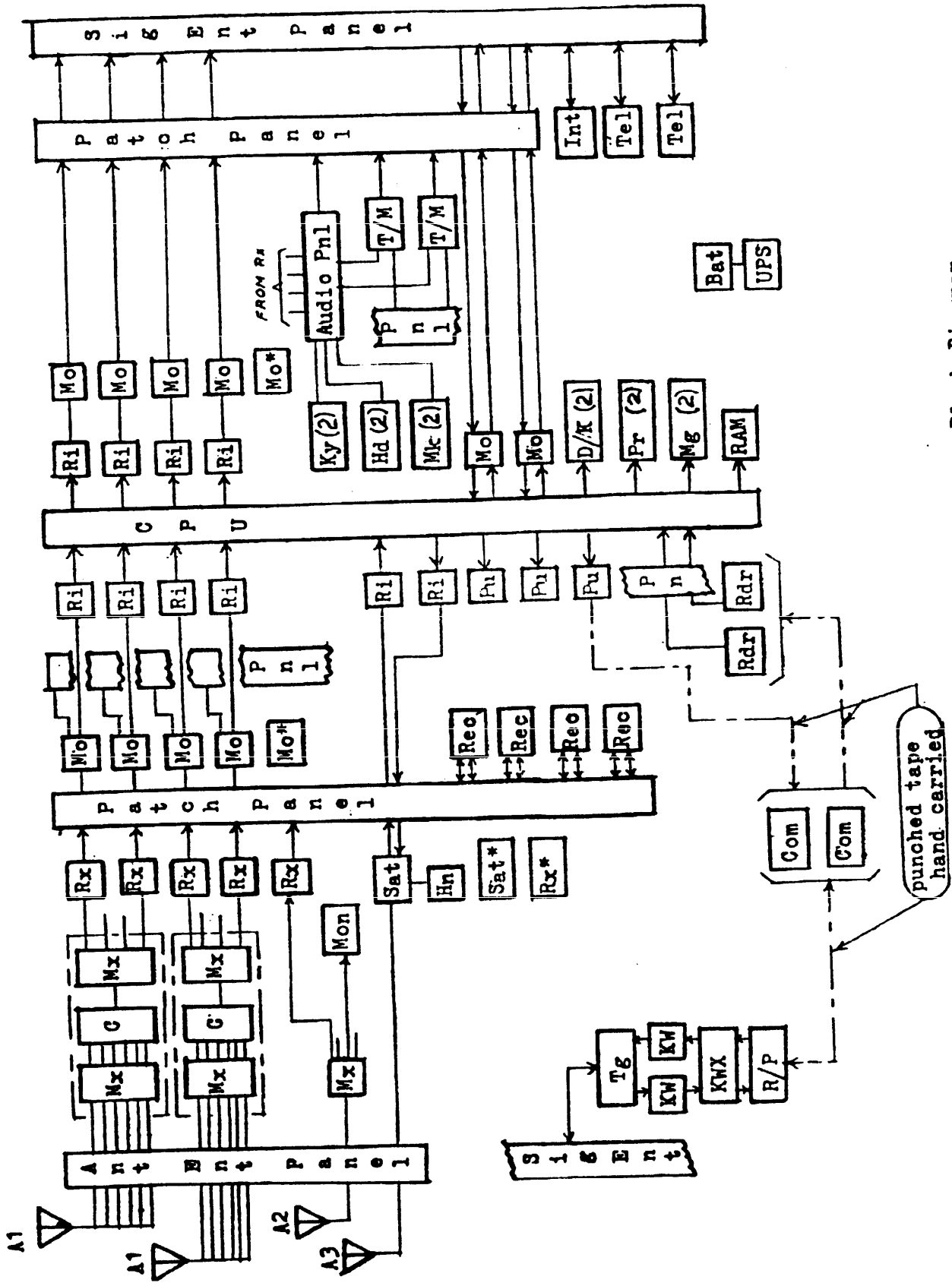


FIGURE 2a. Block Diagram
Receiver Group CR-218/TSC-99

Note: Symbols are given in Fig 2b.

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A1 - - - - Rosette Array Antenna OE-316/TSC-99
A2 - - - - Any other HF antenna
A3 - - - - Satellite radio antenna
Ant Ent Panel - Antenna Entrance Panel on shelter wall
Audio - - - Audio Panel used to monitor audio circuits
Bat - - - Storage battery p/o emergency power system
C - - - Combiner used with rosette array multicouplers
CPU - - - Central Processor Unit
COM - - - COMSEC Terminal
D/K - - - Display-keyboard unit
Hd - - - Headset
Hn - - - Handset p/o satellite radio
Int - - - Intercom used between shelters
Ky - - - Telegraph key for manual CW
KW - - - Coder KW-7/TSEC
KWX - - - Signal Converter KWX-11/TSEC
Mg - - - Magnetic tape read/write unit for CPU
Mk - - - Microphone
Mo - - - Modem (Two types: Message & Processor)
Mon - - - HF Spectrum Monitor
Mx - - - Multicoupler (Two types: Rosette Array use & general purpose)

Pn, Pnl - - p/o patch panel
Pr - - - Page Printer
Pu - - - Paper tape punch
RAM - - - Random access memory
Rdr - - - Paper tape reader
R/P - - - Paper tape reader/punch
Reo - - - Magnetic tape cassette recorder/reproducer for audio system
Ri - - - Radio Interface Unit (Digital Data Comparator Corrector)
Rx = = = HF radio receiver
Sat - - - Satellite radio set
Sig Ent - - Signal Entrance Panel on shelter wall
Tg - - - Wire line modem (Telegraph Terminal TH-22/TG)
T/M - - - Teletype to morse converter
Tel - - - Telephone (Two types: TA-312/PT & TA-938/G)
UPS - - - Uninterruptible power supply

* - - - Identifies a spare item

FIGURE 2b. Symbols Used in Receiver Group Block Diagram .

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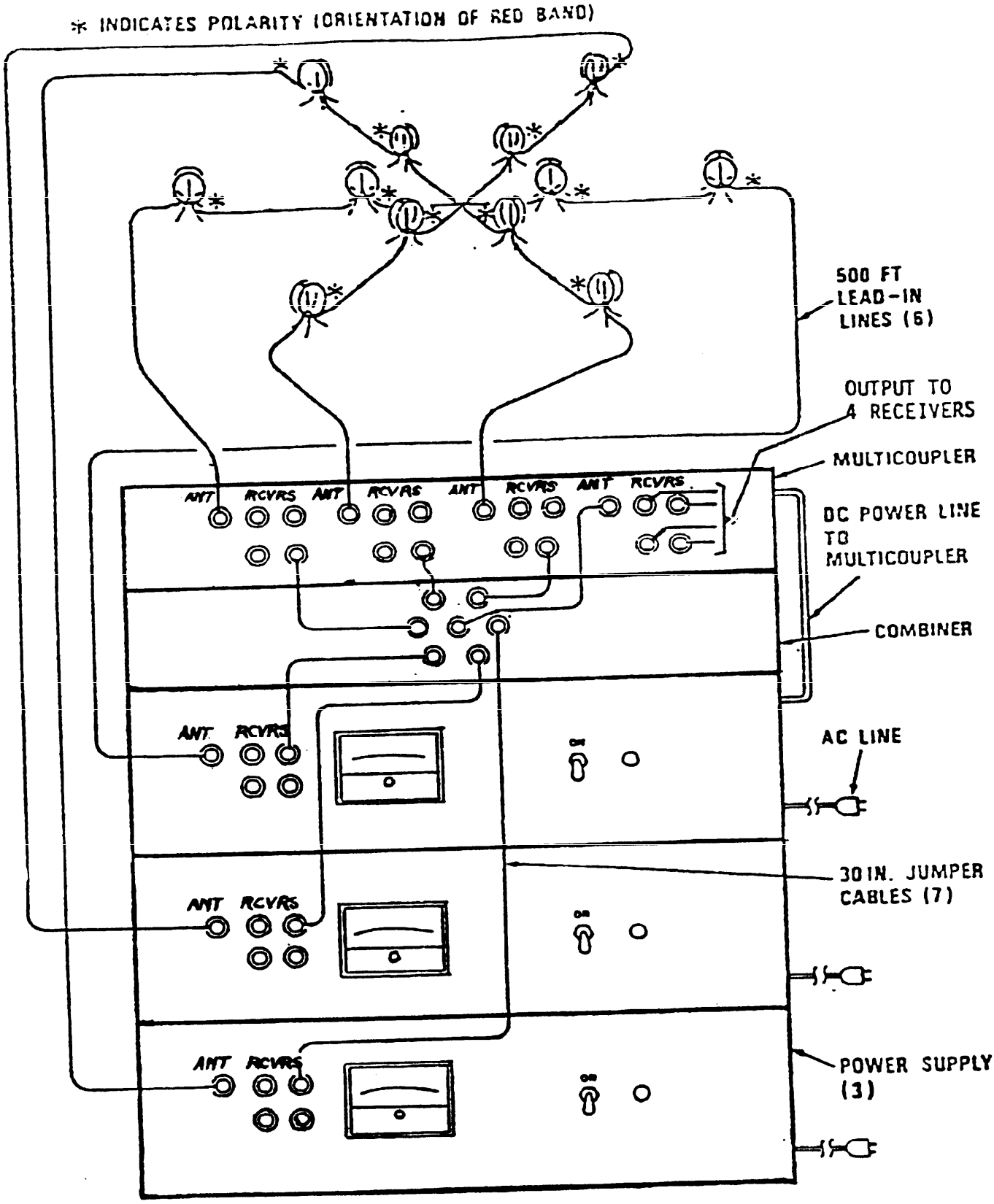


FIGURE 3. Antenna OE-316/TSC-99 Multicouplers Connected Omni-Directional.

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APPENDIX A

Proposed Message and Log Formats

10. Scope This appendix is not a mandatory part of the specification. The information contained herein is intended as a guide for preliminary planning. The actual formats will be governed by the Computer Program Product B-5 Specification which will be prepared for this equipment as a contractual item.

20. Formats

20.1. Proposed Format for Activity Log for Outgoing Radio Messages

NOTE: The number of X's represents the number of digits or characters expected to be used for the entry. Entry will be made by keyboard or by message tape header and printed on the activity log printer.

20.1.1. Initial entry by keyboard

MSG Indent	XXXXXX	Dest Code XXXXX		Send time	XXXXX	Mode	(Burst) - (CW)
Line X	XMTR X	FREQ XXXXX	PWR XX	ANT X			(Voice)
Line X	XMTR Y	FREQ YYYYY	PWR XX	ANT Y			

20.1.2. Automatic printout after message sent

MSG XXXXX TIME XXXX

20.1.3. Remarks: (Keyboard entry)

Example: XMTR X OUT OF SERVICE TIME XXXX
Example: XMTR X SERVICE RESTORED TIME XXXX
Example: WIRE LINE X OUT OF SERVICE TIME XXXX

20.2. Proposed Stored Message Summary

Date Time Group	Msg Ident	Status
XXXXXX	XXXXX	(Loaded, Punched, Scheduled, or Complete)

20.3. Proposed Daily Summary of Outgoing Radio Message Traffic

Heading (Containing address in military format)

Date Time Group	Msg Ident
XXXXXX	XXXXX

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20.4. Proposed Format for Incoming Radio Message Log

TIME RCVD	MSG IDENT	SEND STATION	READ-ABILTIY	FREQ USED	GROUP COUNT	MODE
XXXX	XXXXXX	XXXXX	X	XXXXX	XXX	(Burst, CW, or voice)

REMARKS: (will be on line following the data, if any)

NOTE: Some of the above information will have to be provided by the receiver operator viewing the message as it arrives and by the COMSEC operator after processing.

20.5. Proposed Format for Daily Summary of Incoming Traffic

Heading (Containing address in military form)

Date Time Group	Msg Indent	Station	Readibility	Group Count
XXXXXX	XXXXXX	XXXXX	X	XX

20.6. Proposed Daily Equipment Status Report

TRANSMITTERS	XXXX
HF RECEIVERS	XXXX
SATELLITE TERMINAL	XXXX
PROCESSOR	XXXX
COMSEC	XXXX
ANTENNA	XXXX

Note: Entry is in the form of "well" or "sick"

REMARKS: This will list equipment outage, if any, and the duration.

20.7. Proposed Monthly HF Frequency Prediction Format

This data will be delivered by teletype in the format shown.

Prop Data for Station XXXXX
For (Month)

ZT	MUF	FOT	LUF
00	XX.X	XX.X	XX.X
02	XX.X	XX.X	XX.X
04	XX.X	XX.X	XX.X
06	XX.X	XX.X	XX.X
08	XX.X	XX.X	XX.X
10	XX.x	XX.X	XX.X
12	XX.x	XX.X	XX.X
14	XX.x	XX.X	XX.X
16	XX.X	XX.X	XX.X
18	XX.X	XX.X	XX.X
20	XX.X	XX.X	XX.X
22	XX.X	XX.X	XX.X

Note: There will be a maximum of 36 tables, one for each outstation.

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20.8. Proposed "Frequency Used" File

OUTSTATION IDENT XXXXX

FREQ	MONTH/DAY-HOUR
XX.XXXX	XX/XX-XX

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APPENDIX B

Storage Space Requirements

10. Scope: This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

20. Storage Space The space requirement data given below is that required at the COMSEC operator's position on the basis of one half to a full day supply. (REF. 3.10.4).

Pads	6-1/2 lg x 4w x 6 h
Tapes	3-1/2 lg x 3-1/2 w x 9 h
Teletype tape	8 lg x 8 w x 6 h
Writing pads	10-1/2 lg x 8 w x 2 h
Pencil box	7-1/2 lg x 2 w x 1 h

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-49187	Communications Central AN/TSC-99

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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